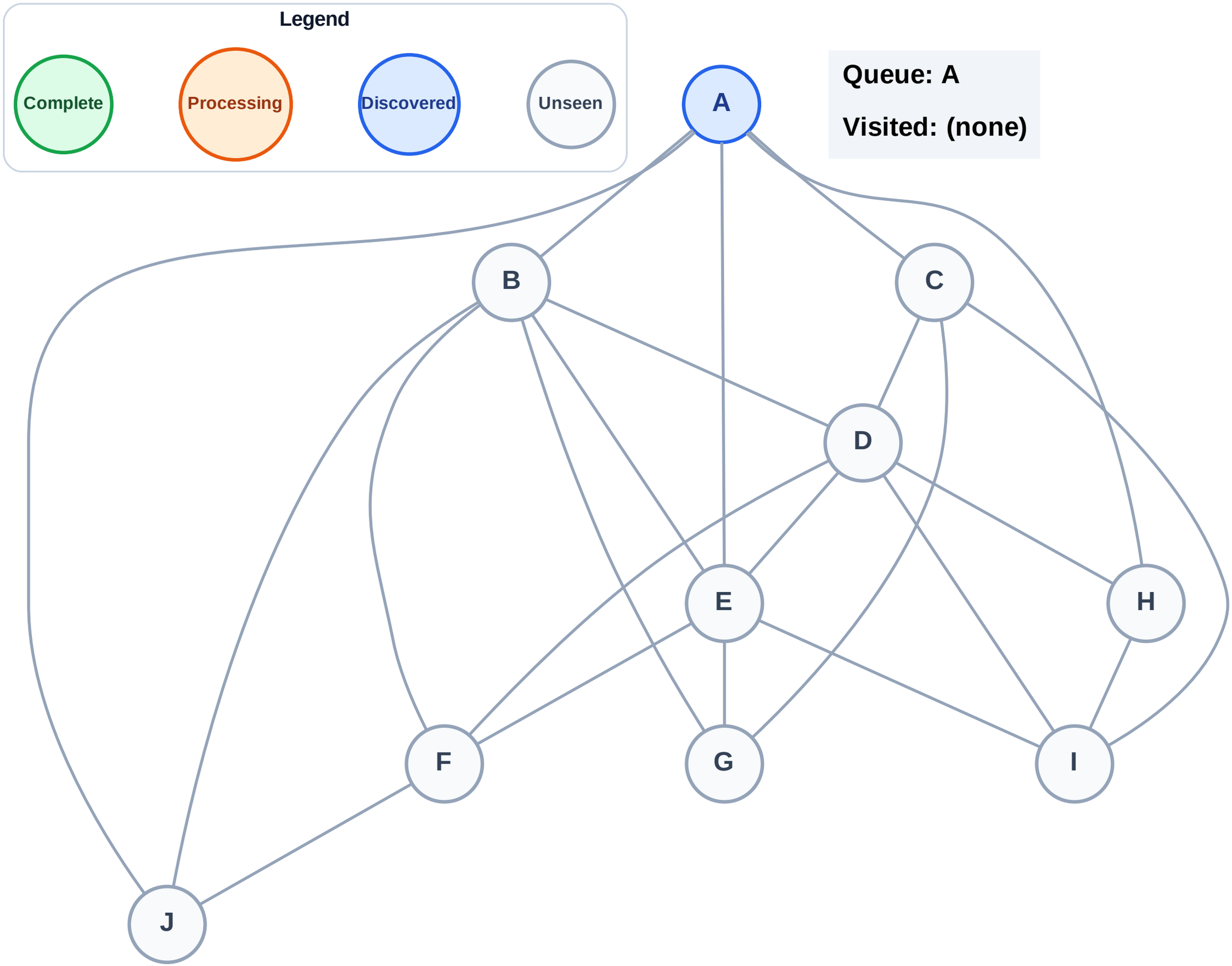
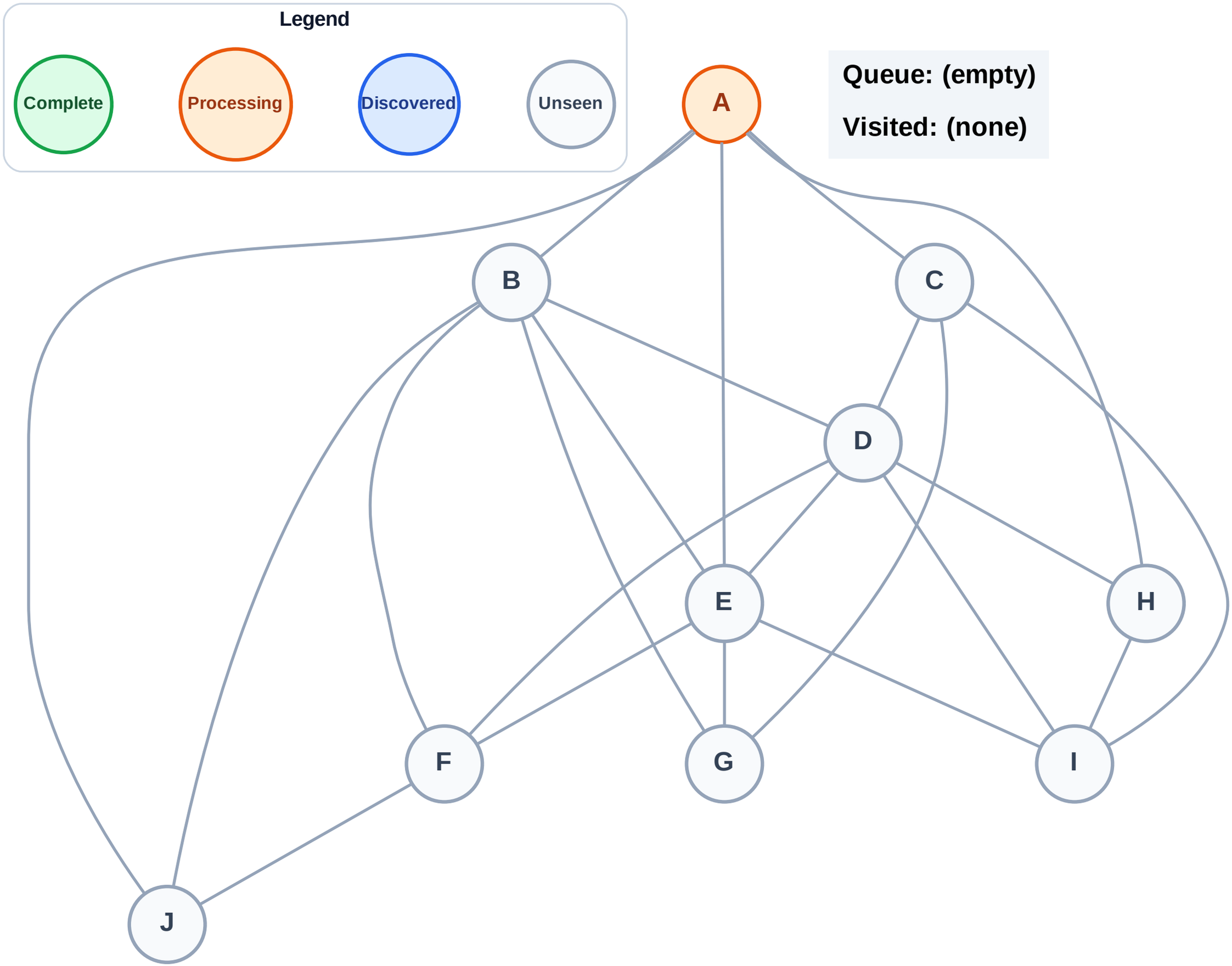
BFS Traversal

Step 1: Start at A



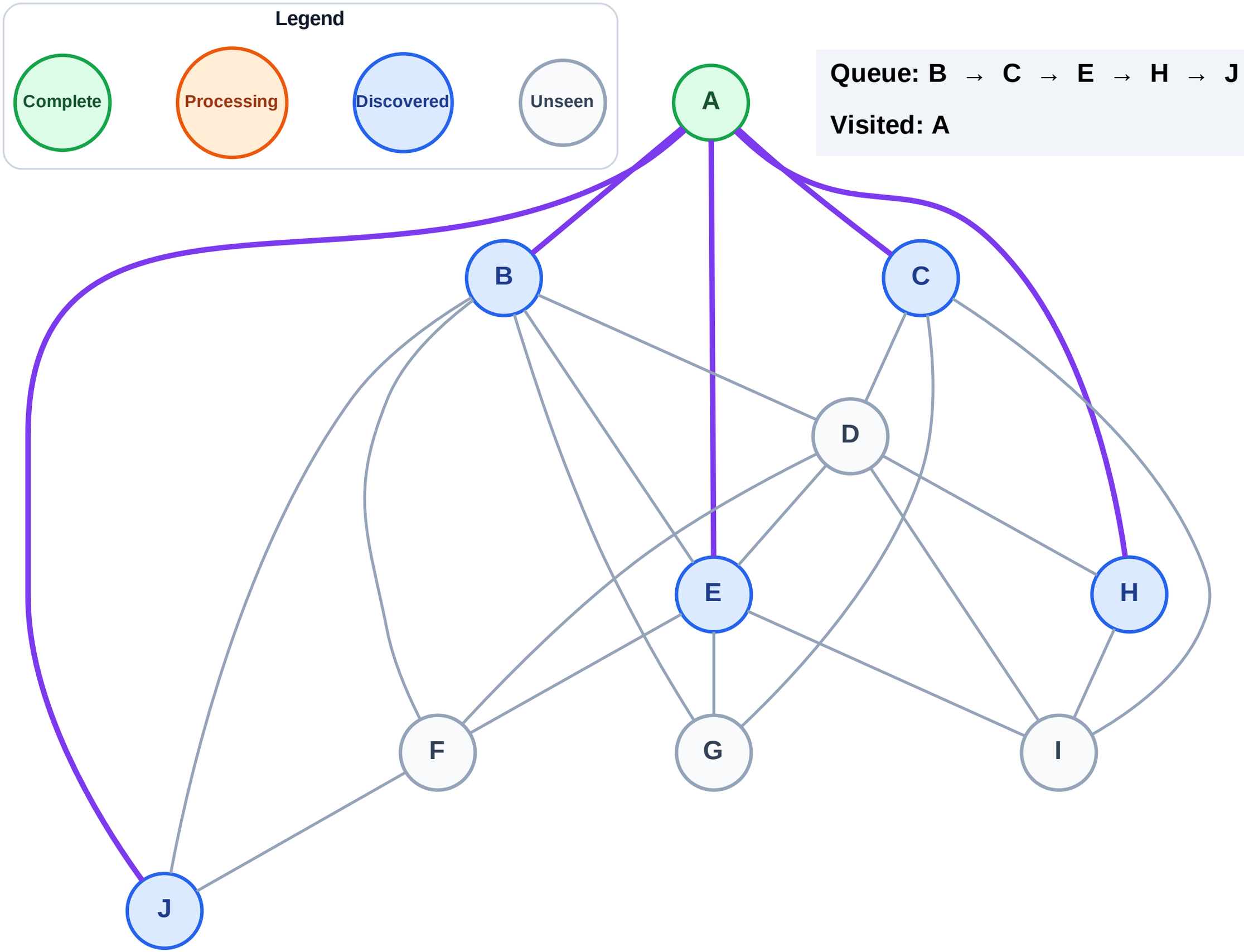
BFS Traversal

Step 2: Process node A



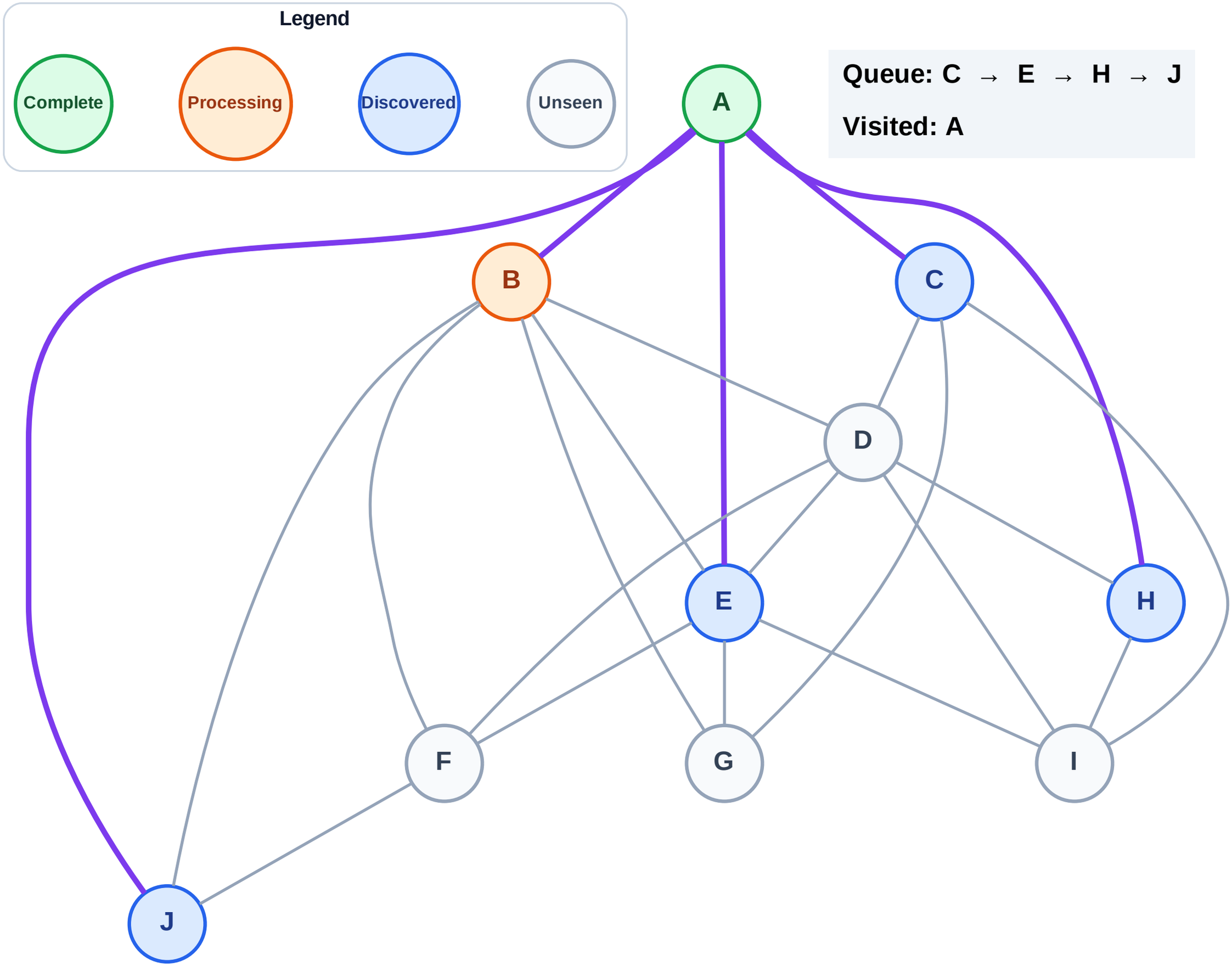
BFS Traversal

Step 3: Discover B, C, E, H, J from A



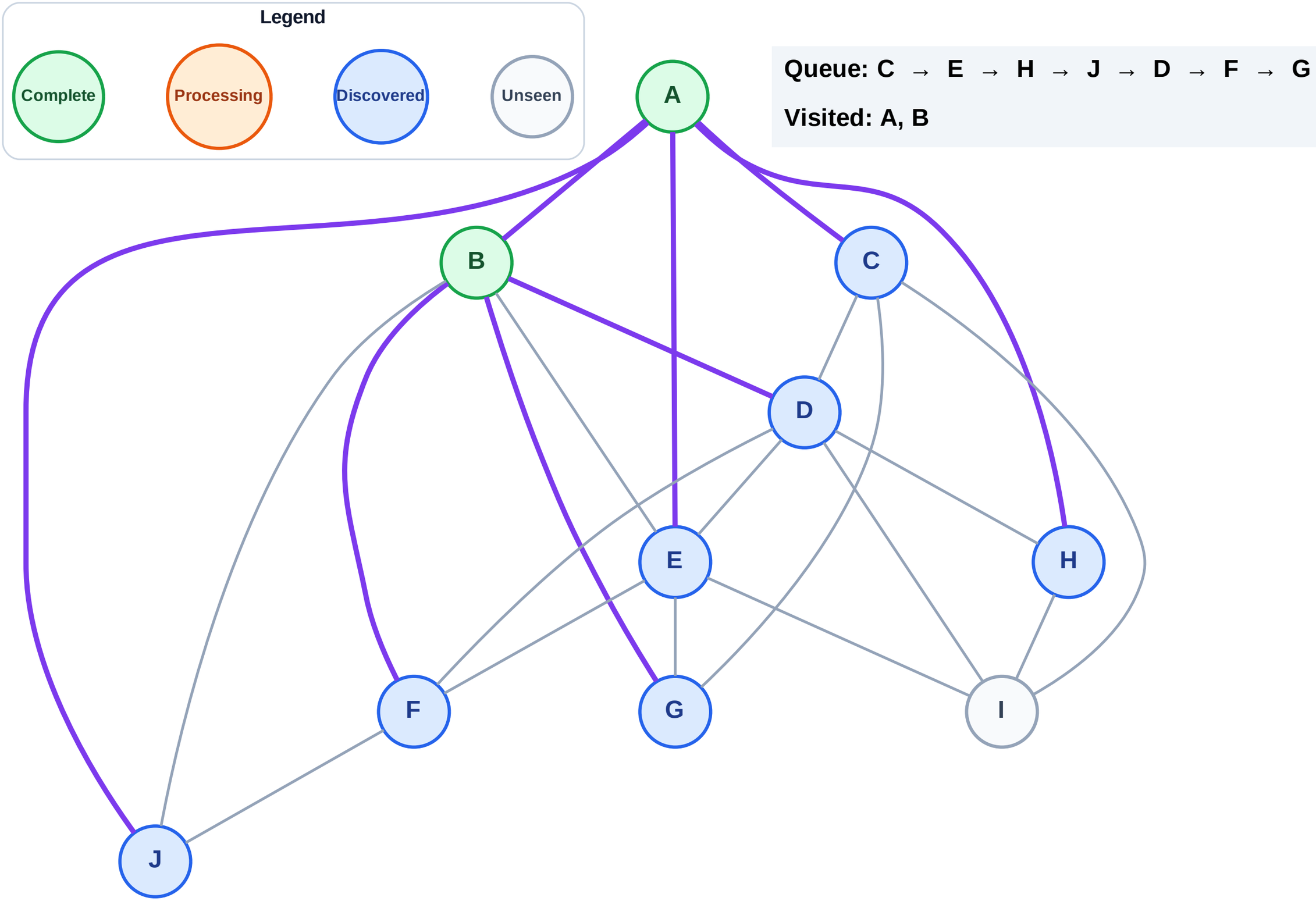
BFS Traversal

Step 4: Process node B



BFS Traversal

Step 5: Discover D, F, G from B

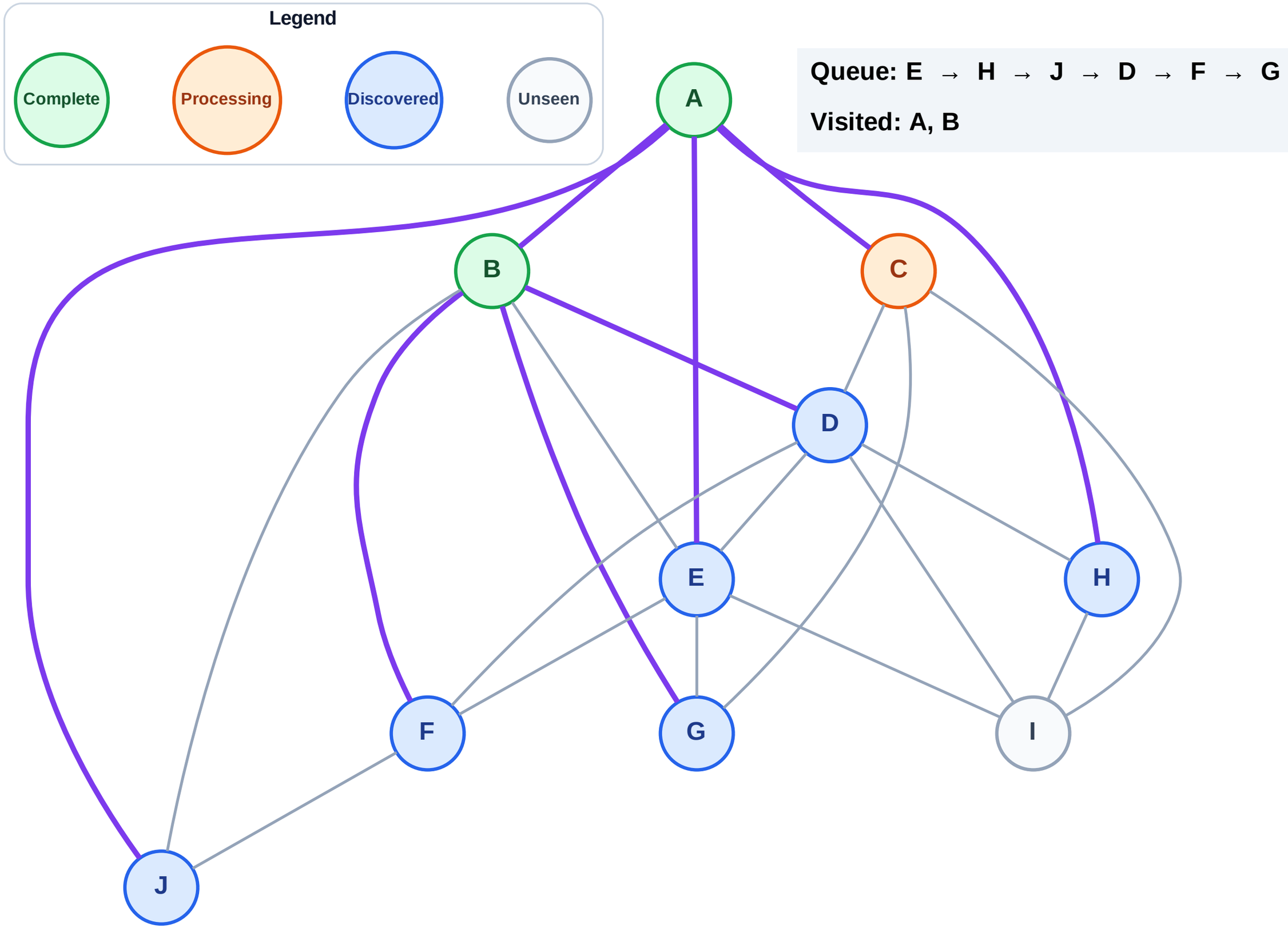


Queue: C → E → H → J → D → F → G

Visited: A, B

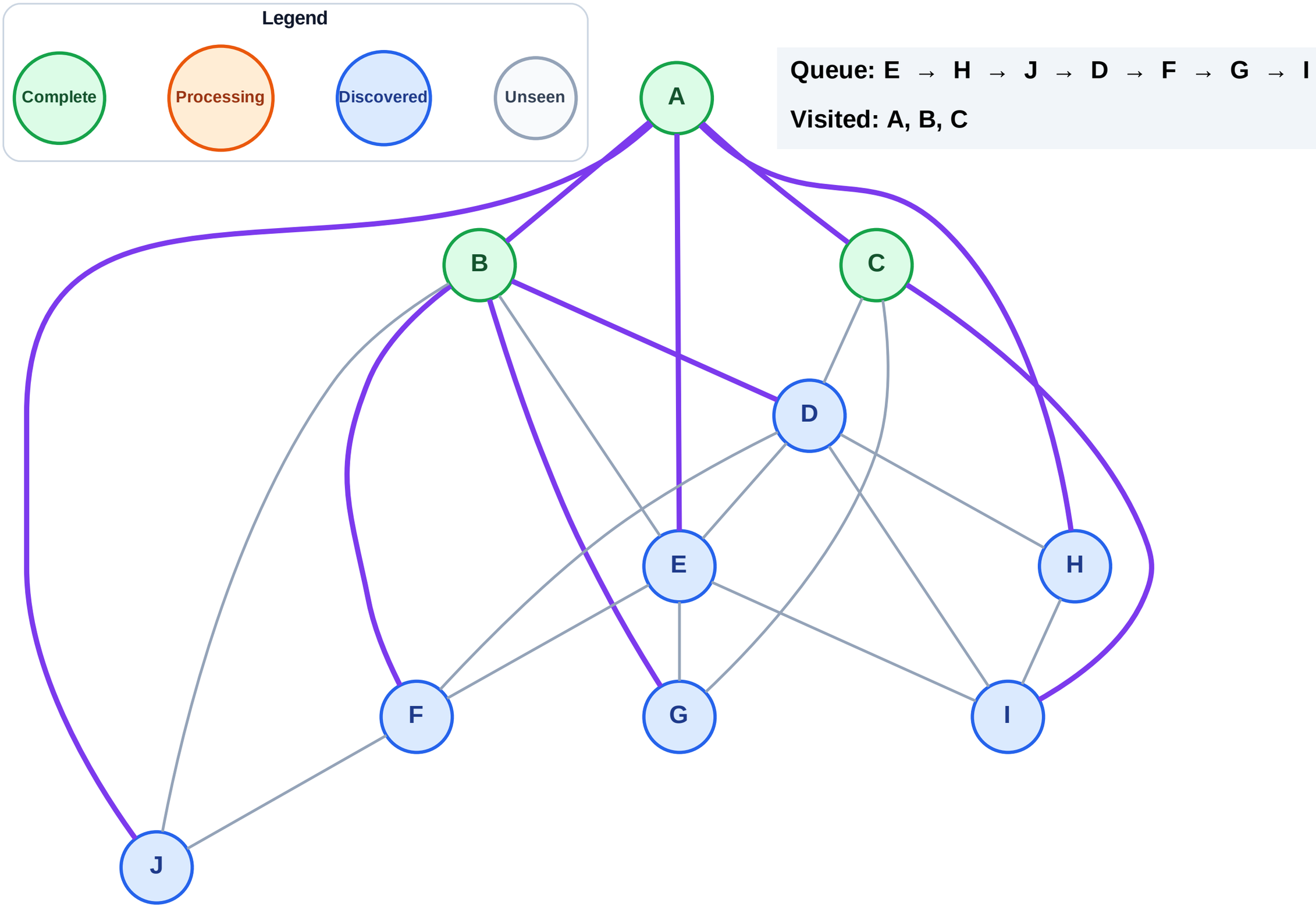
BFS Traversal

Step 6: Process node C



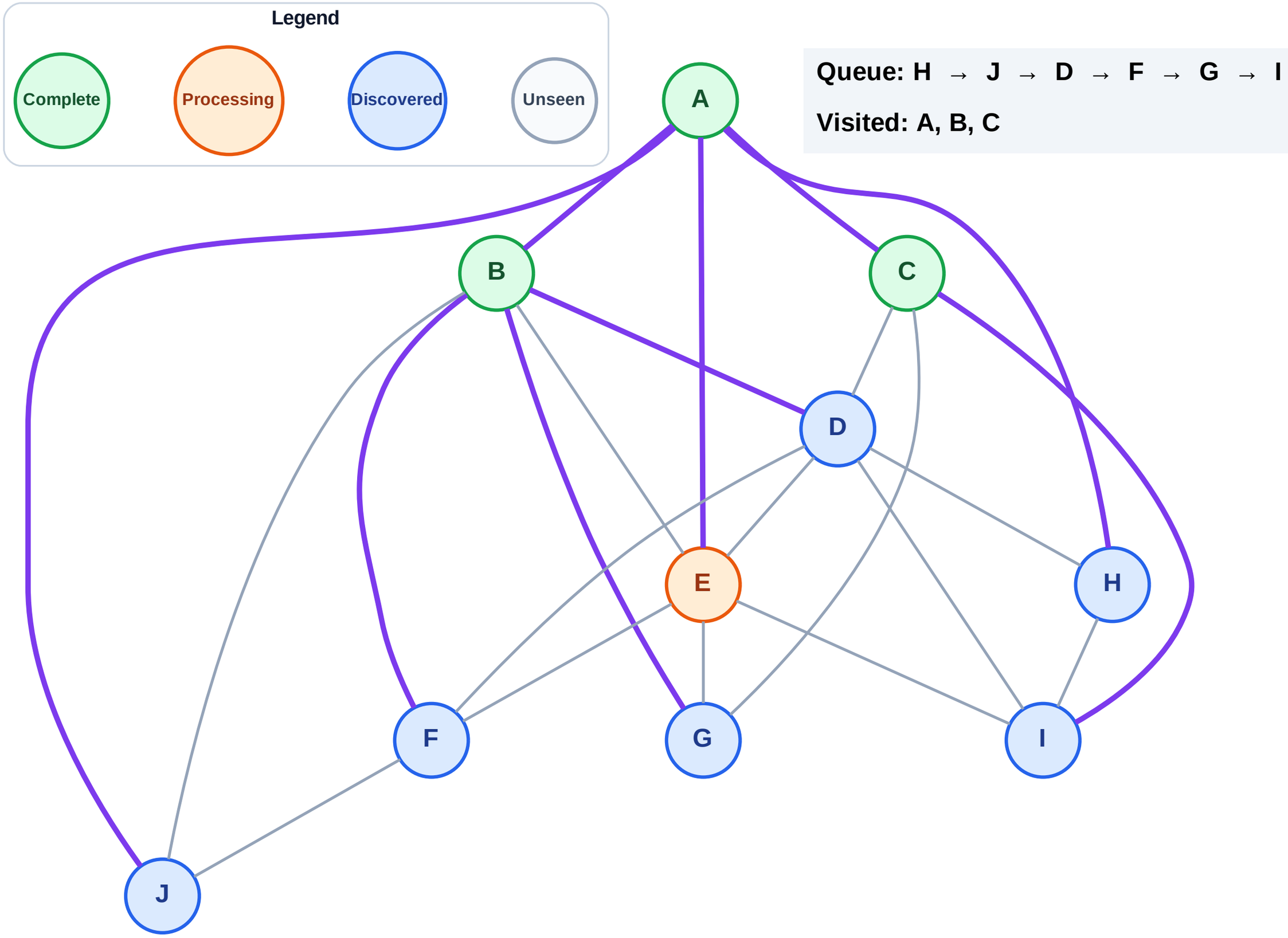
BFS Traversal

Step 7: Discover I from C



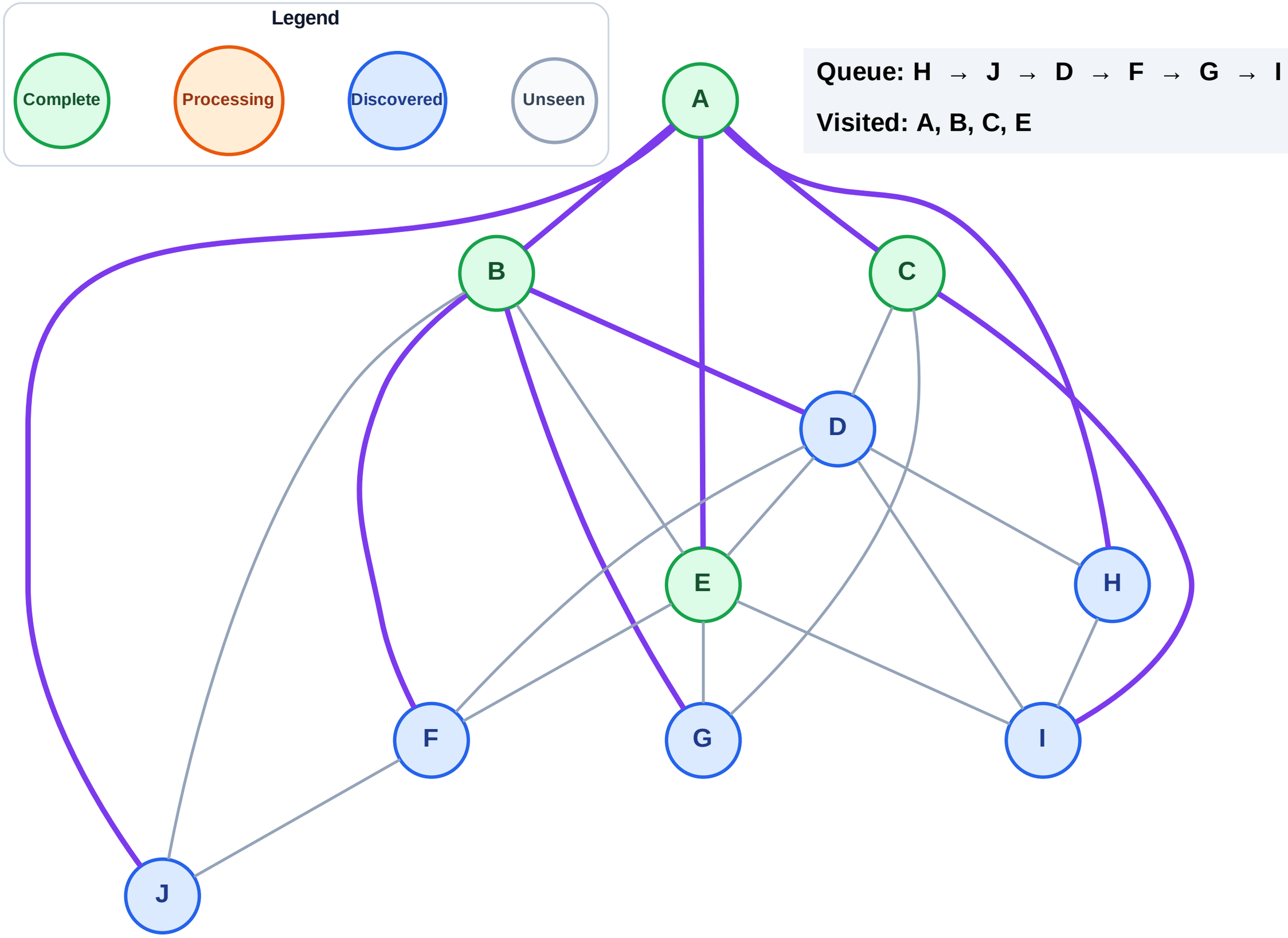
BFS Traversal

Step 8: Process node E



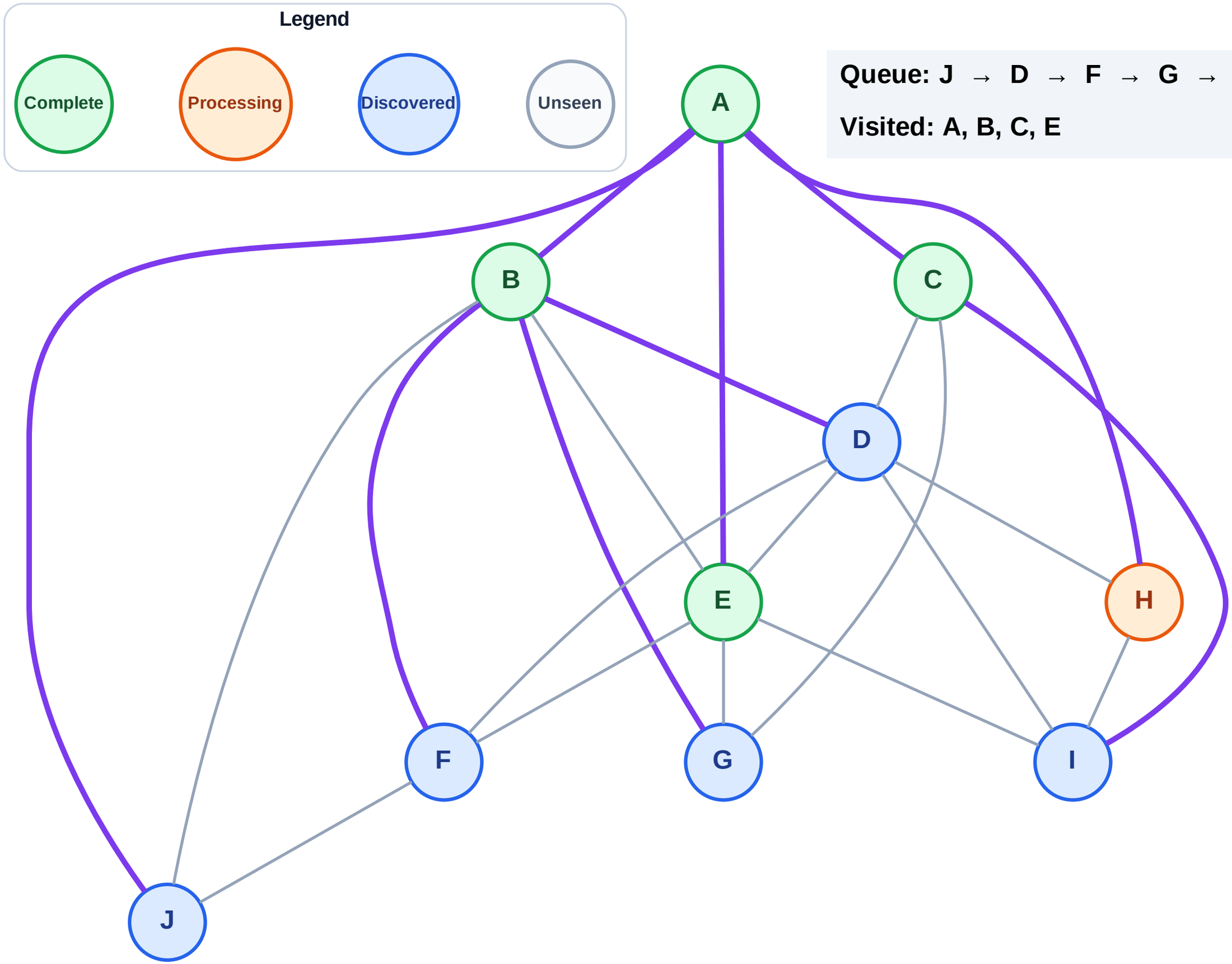
BFS Traversal

Step 9: No new neighbors from E



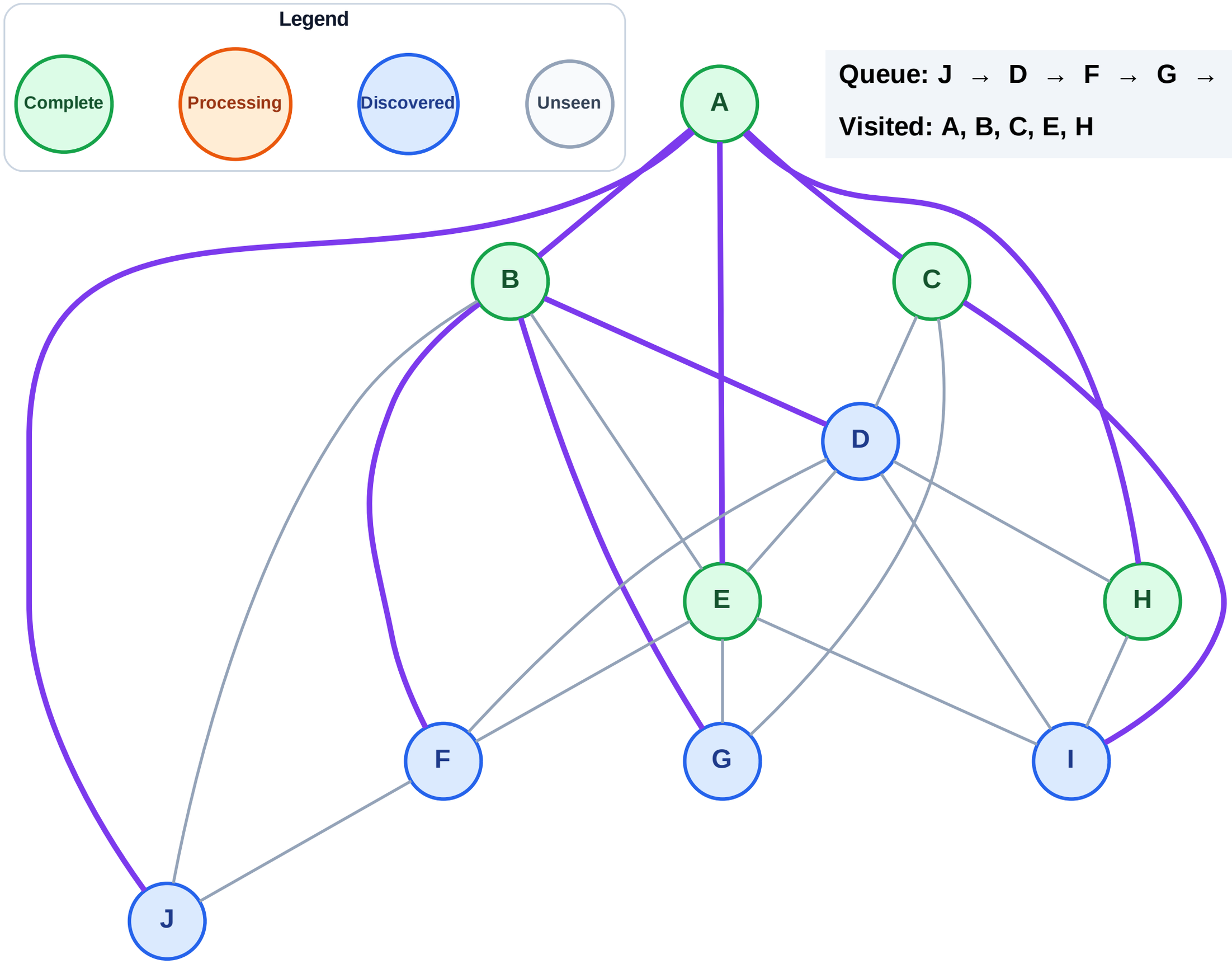
BFS Traversal

Step 10: Process node H



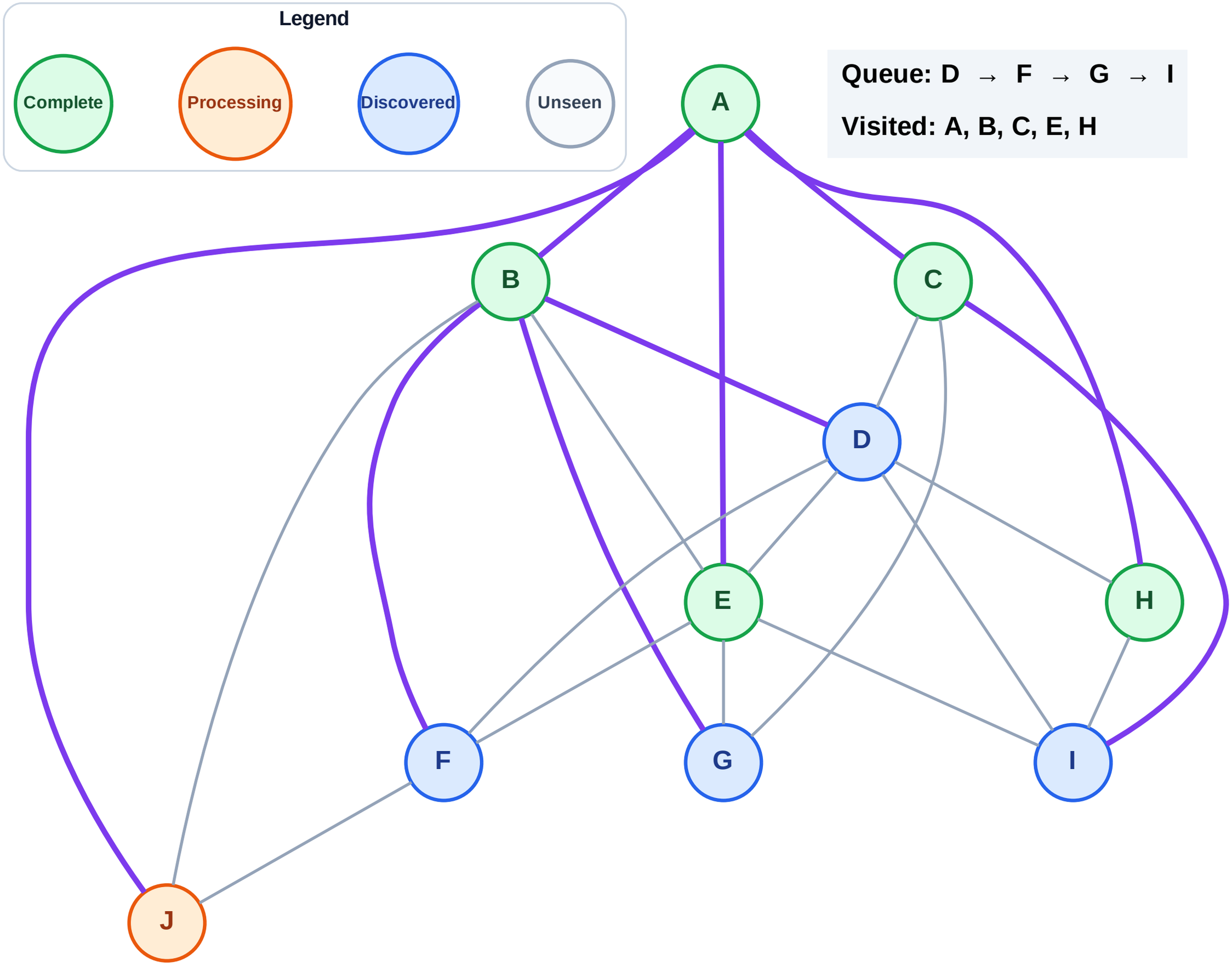
BFS Traversal

Step 11: No new neighbors from H



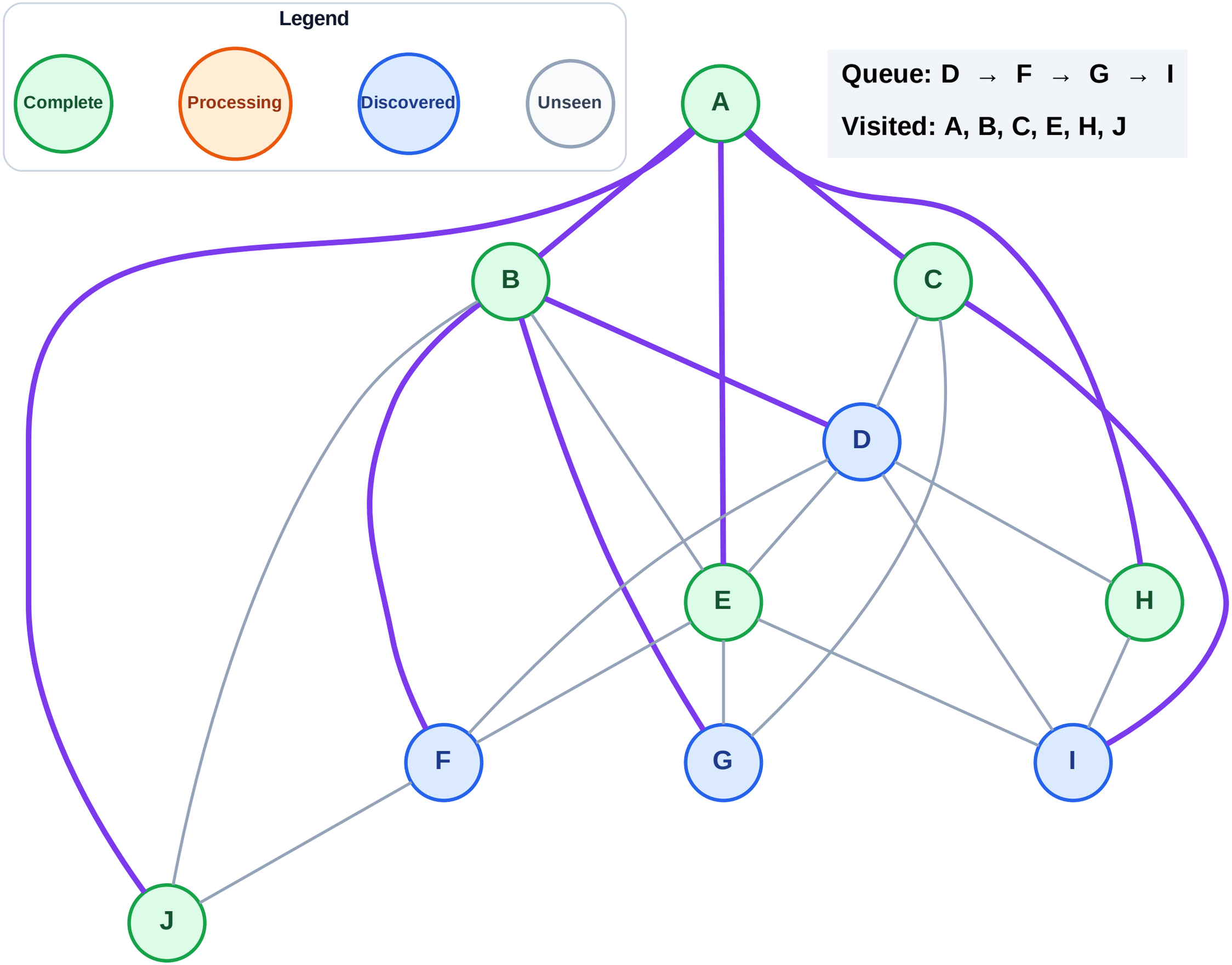
BFS Traversal

Step 12: Process node J



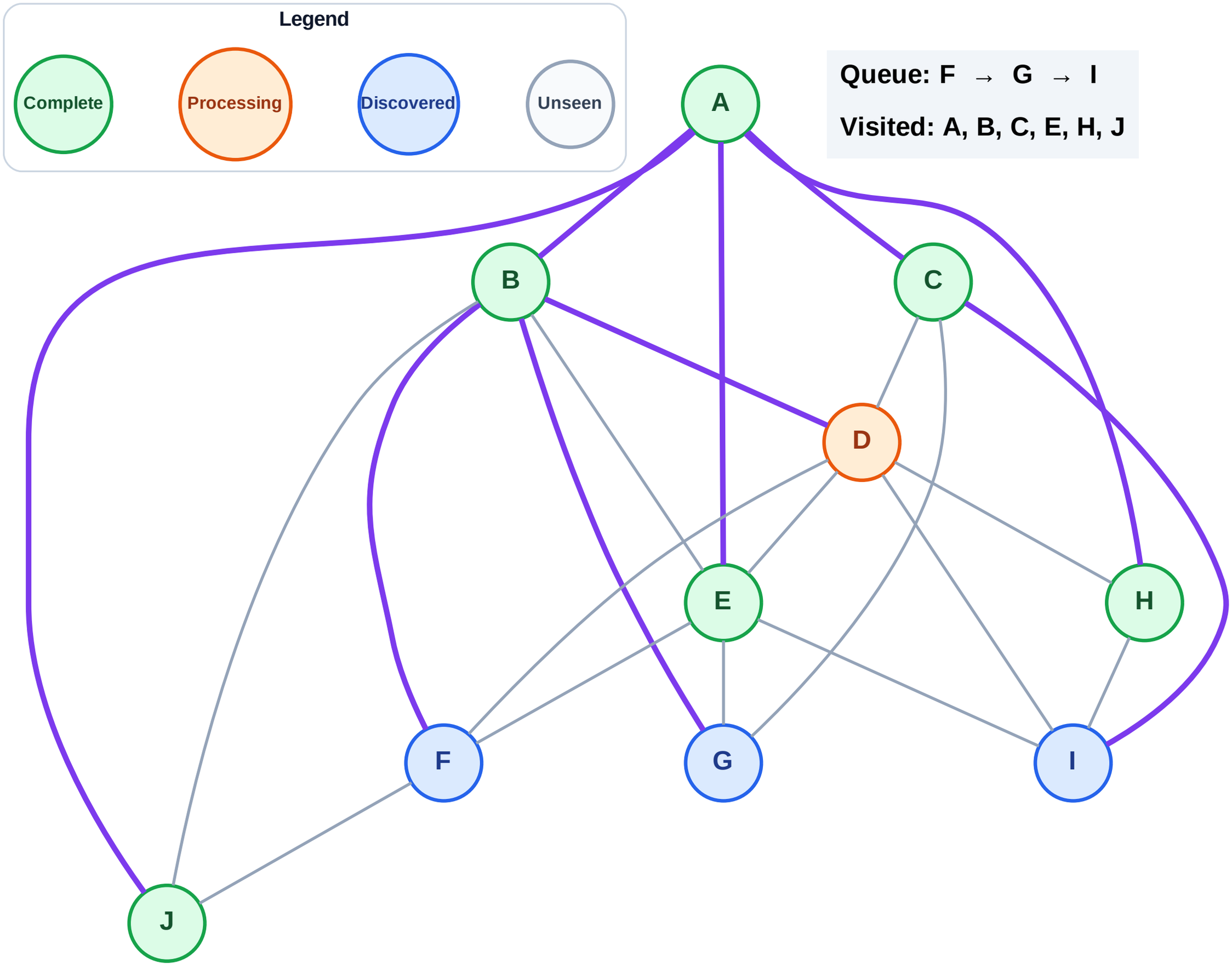
BFS Traversal

Step 13: No new neighbors from J



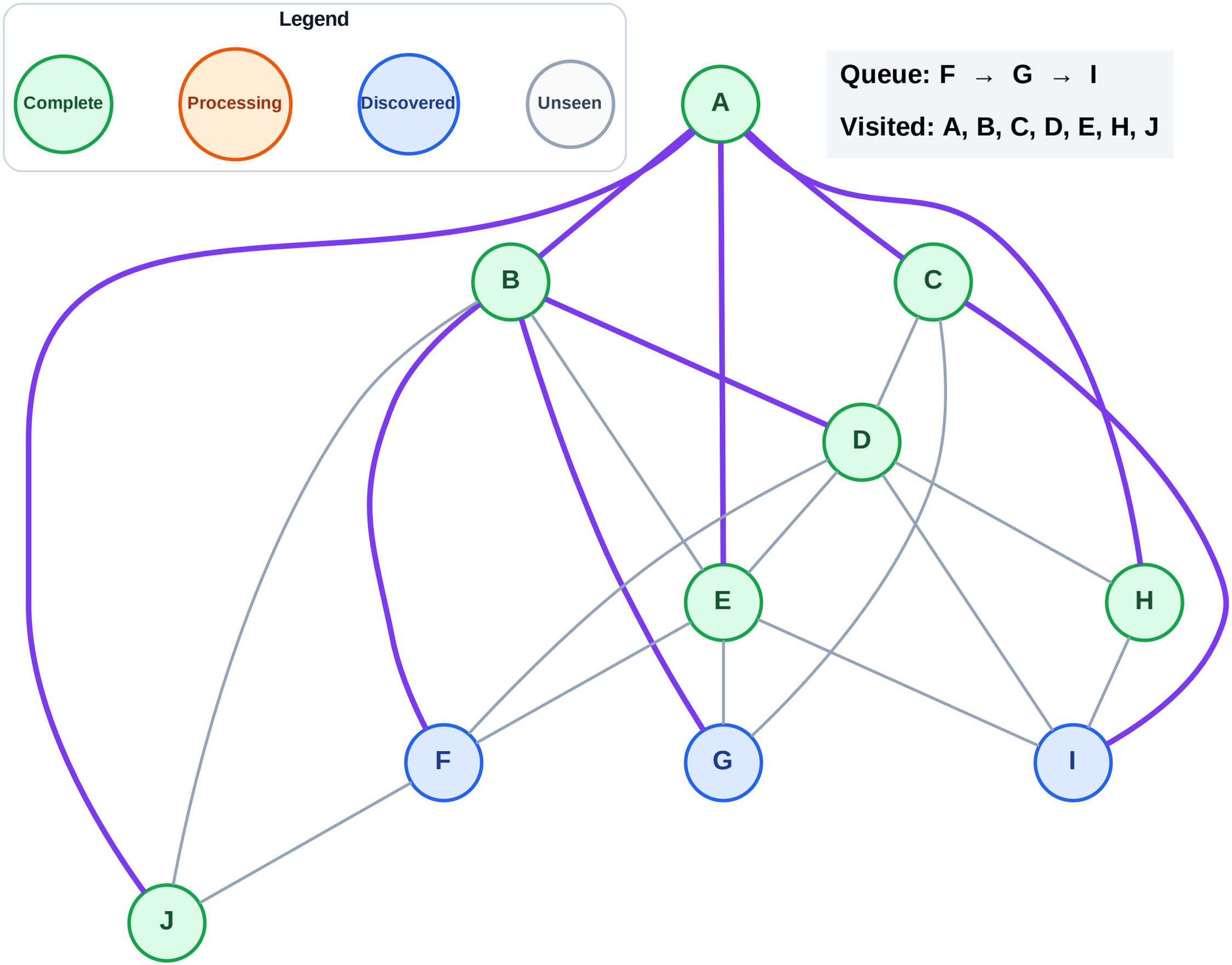
BFS Traversal

Step 14: Process node D



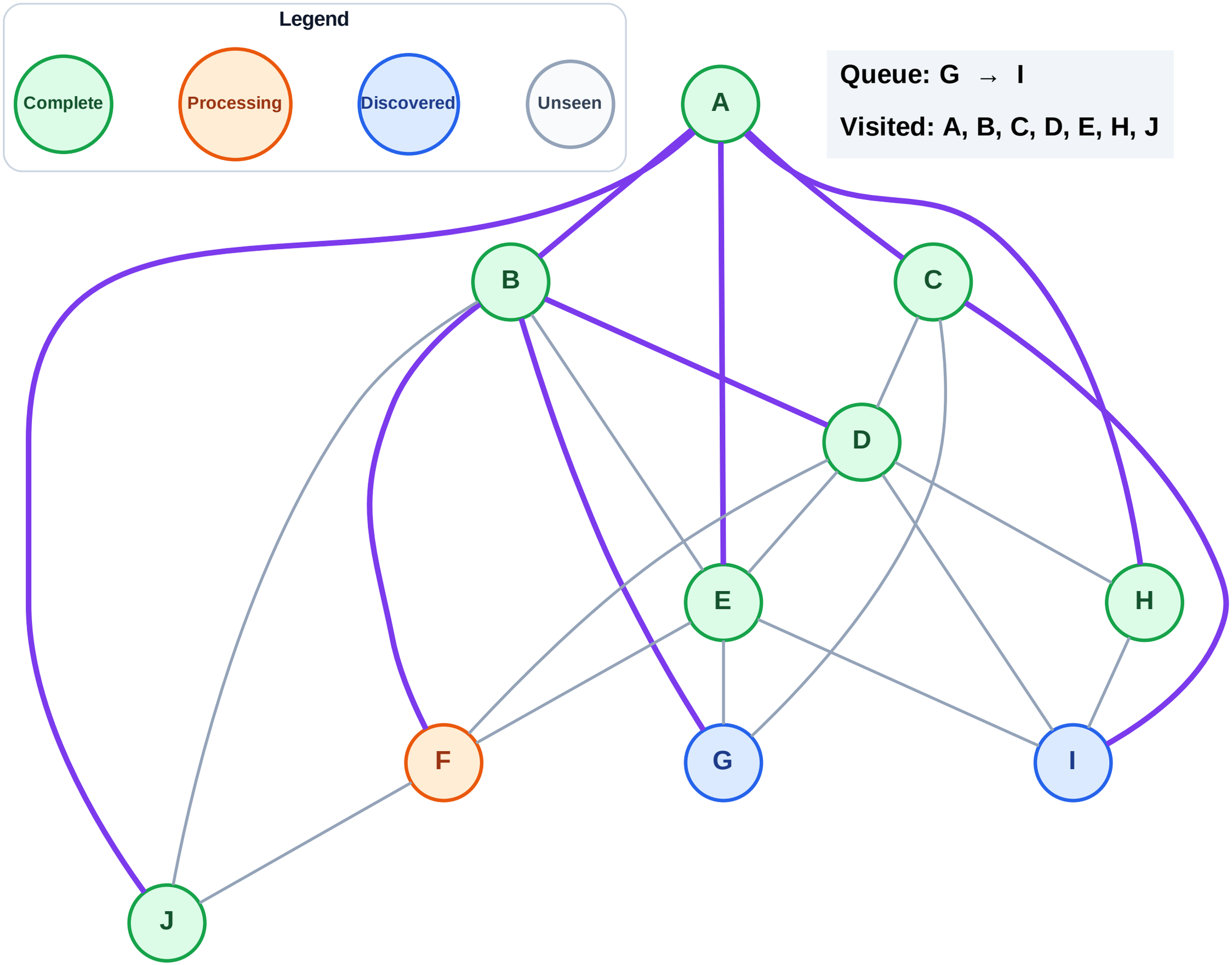
BFS Traversal

Step 15: No new neighbors from D



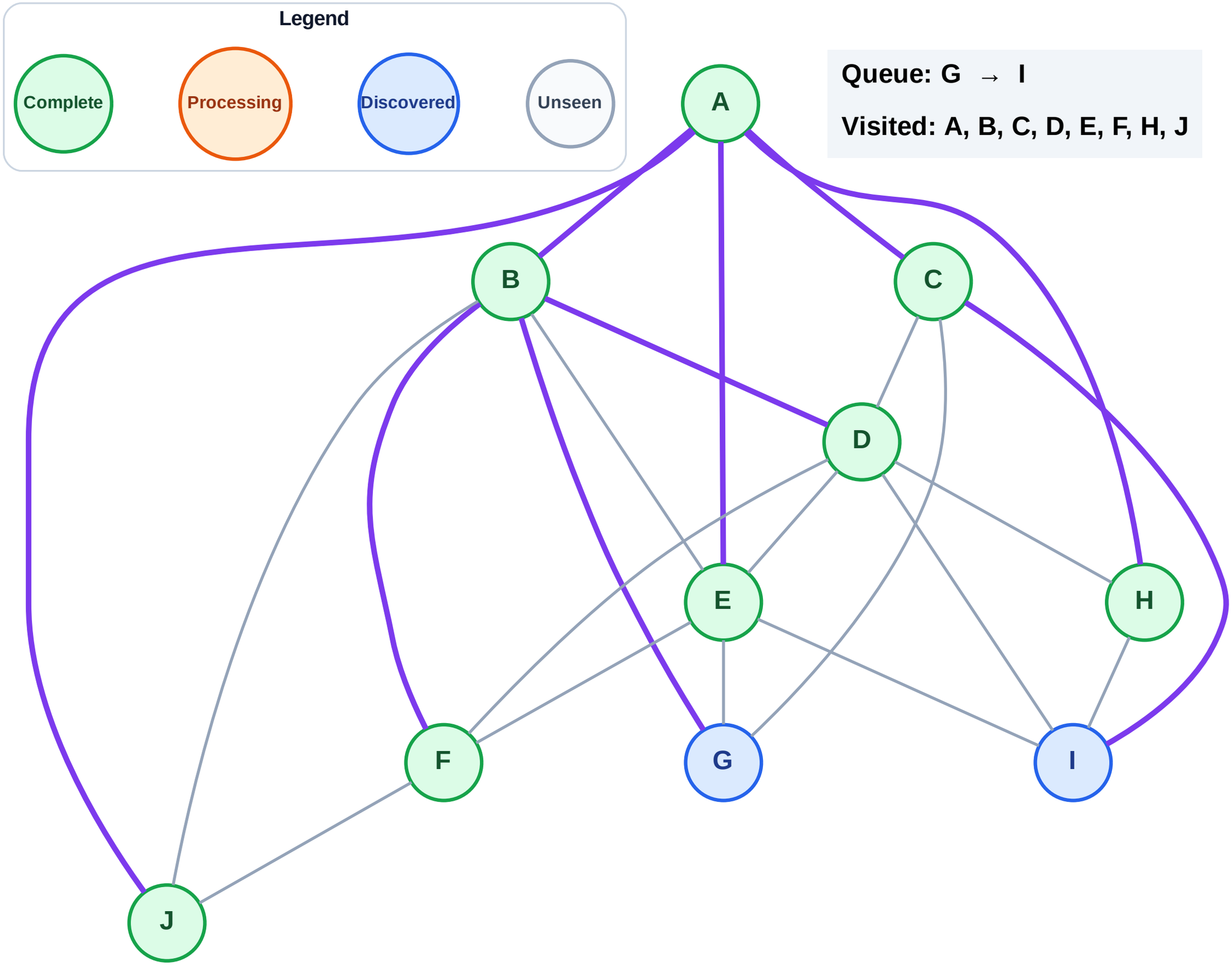
BFS Traversal

Step 16: Process node F



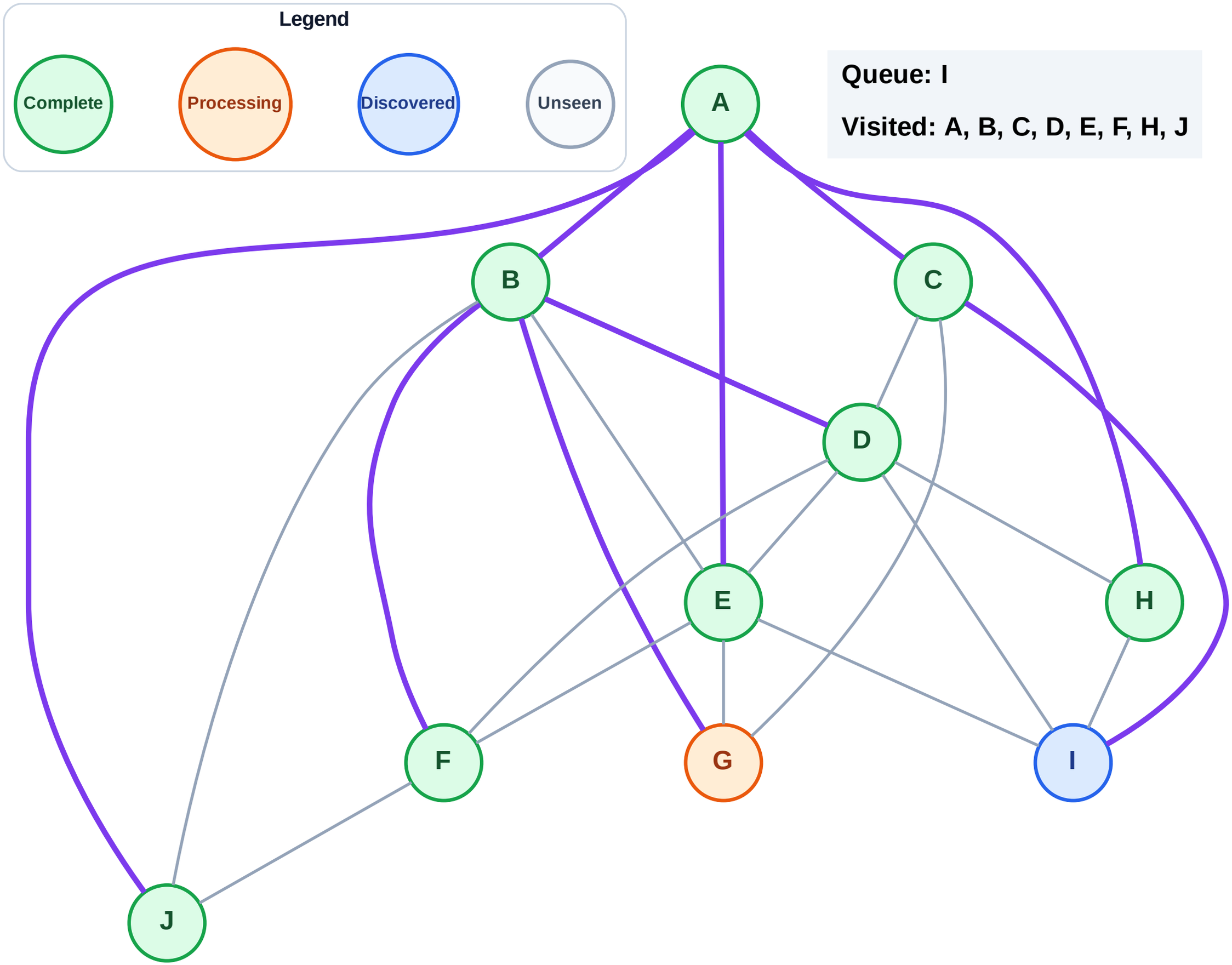
BFS Traversal

Step 17: No new neighbors from F



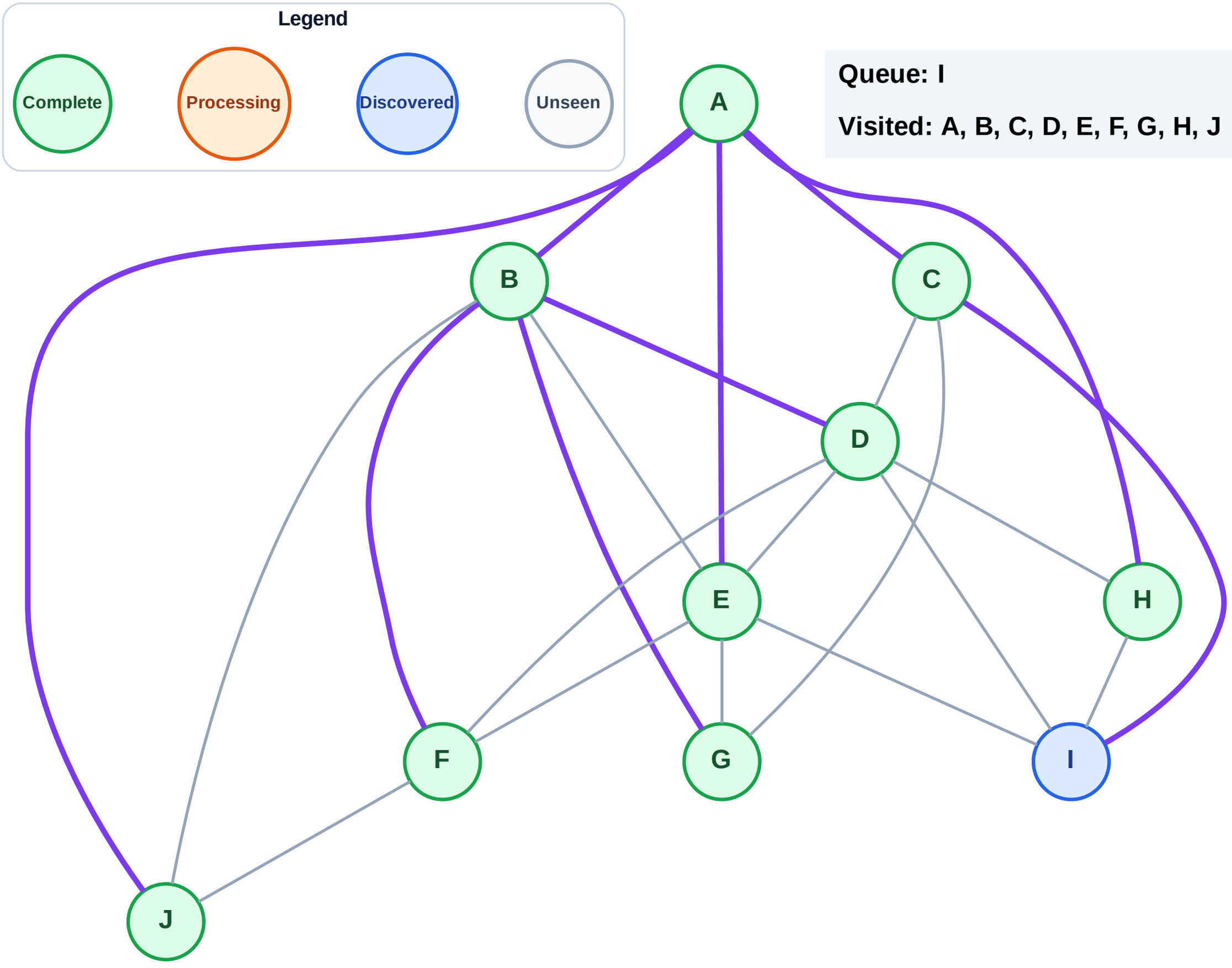
BFS Traversal

Step 18: Process node G



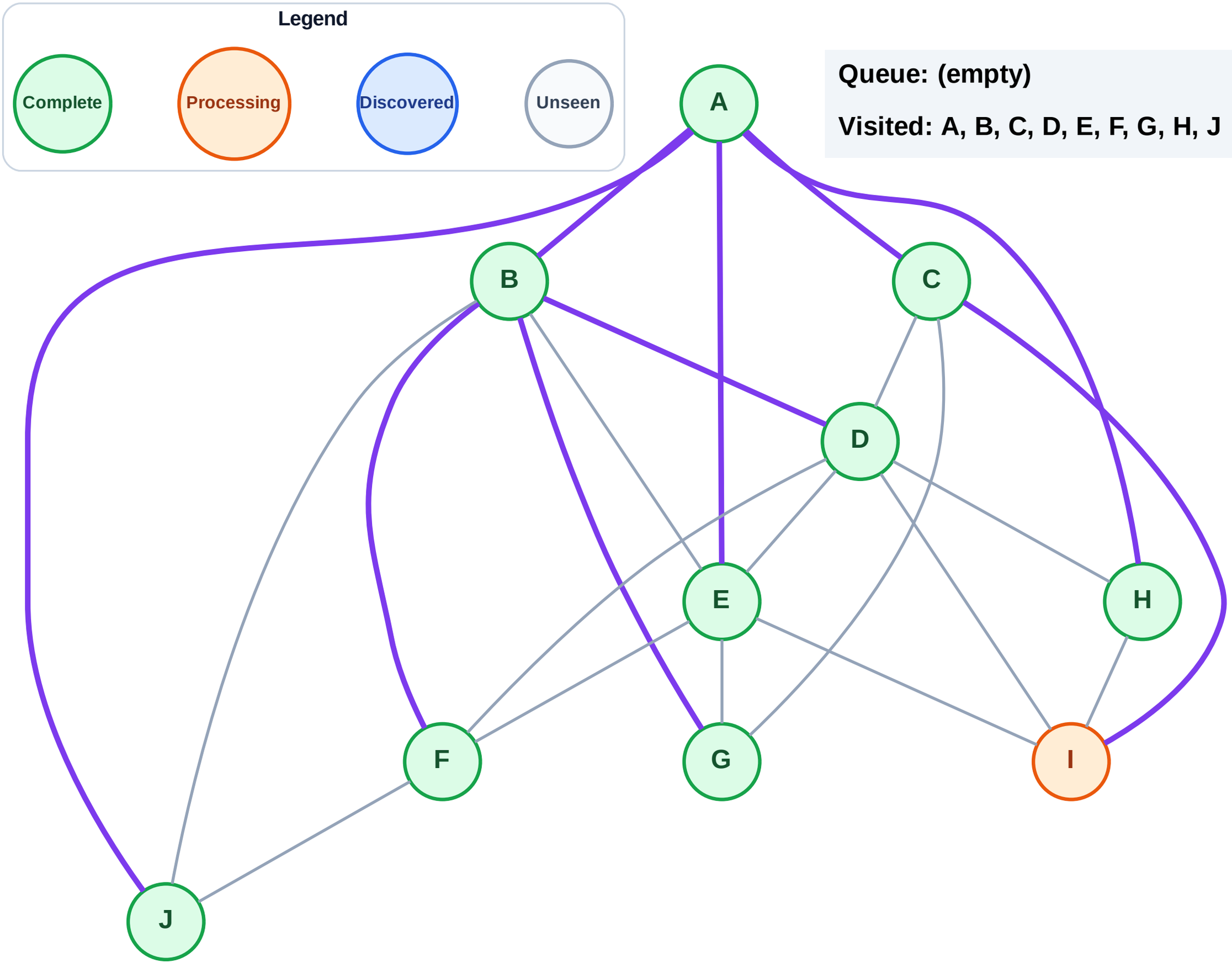
BFS Traversal

Step 19: No new neighbors from G



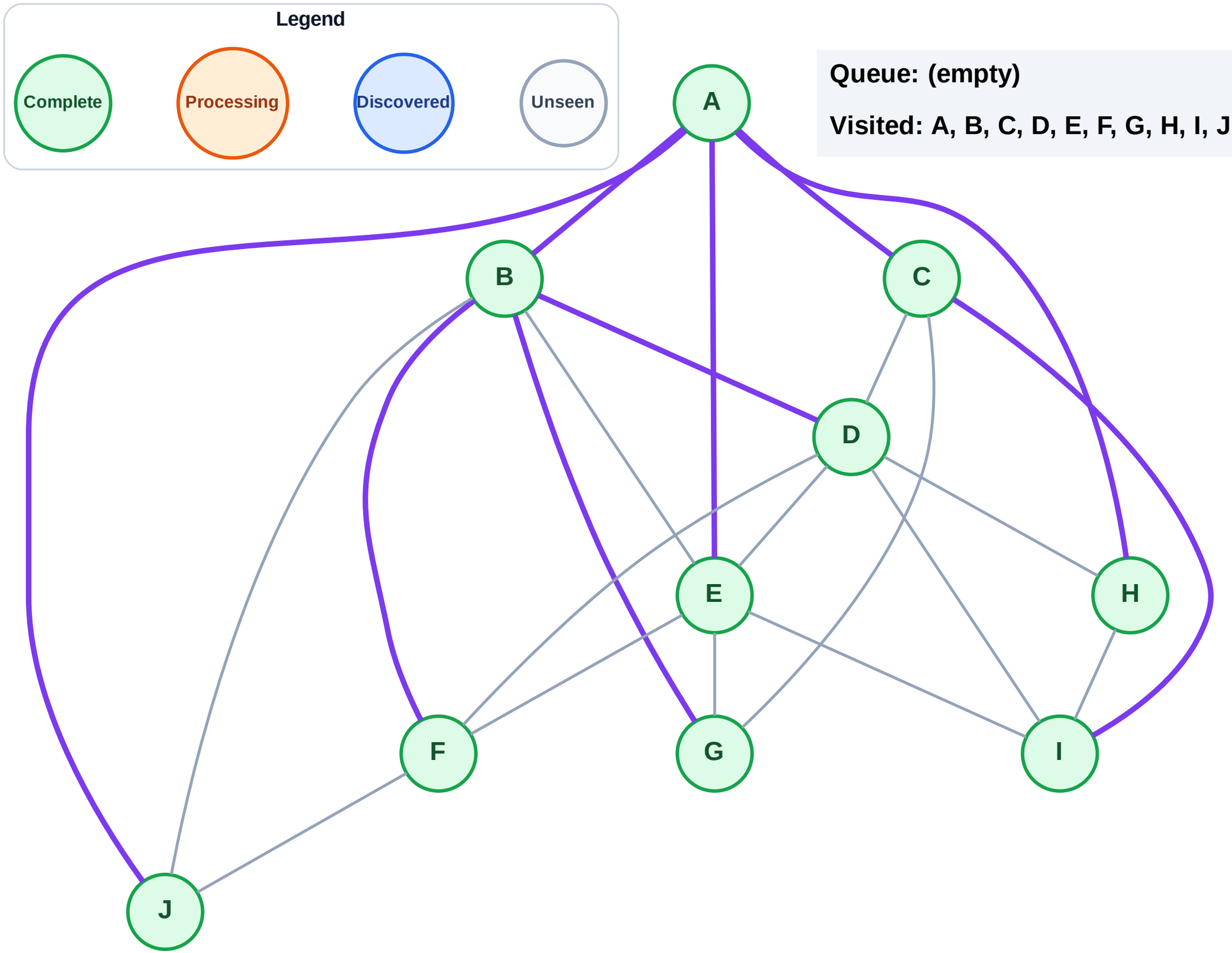
BFS Traversal

Step 20: Process node I



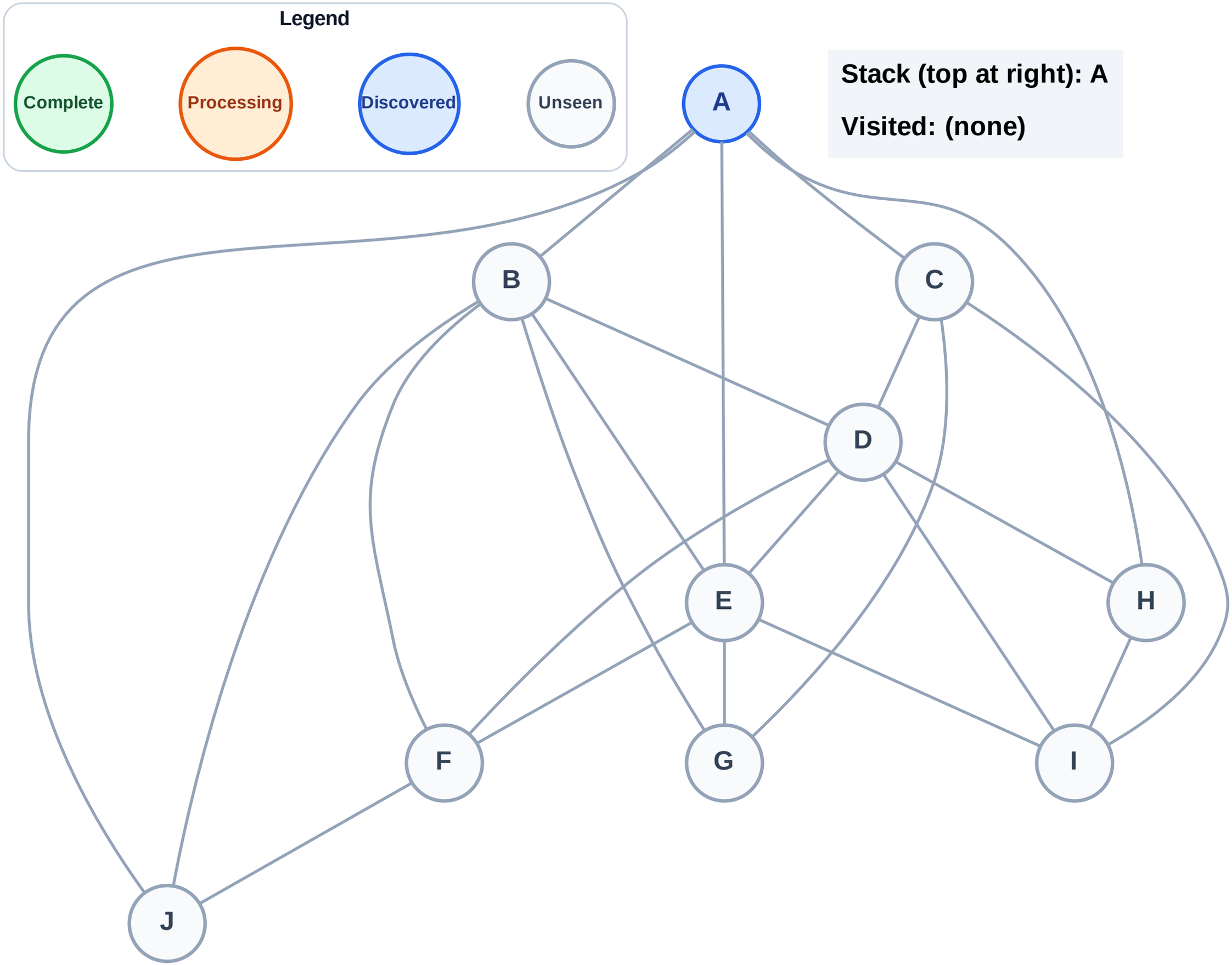
BFS Traversal

Step 21: No new neighbors from I



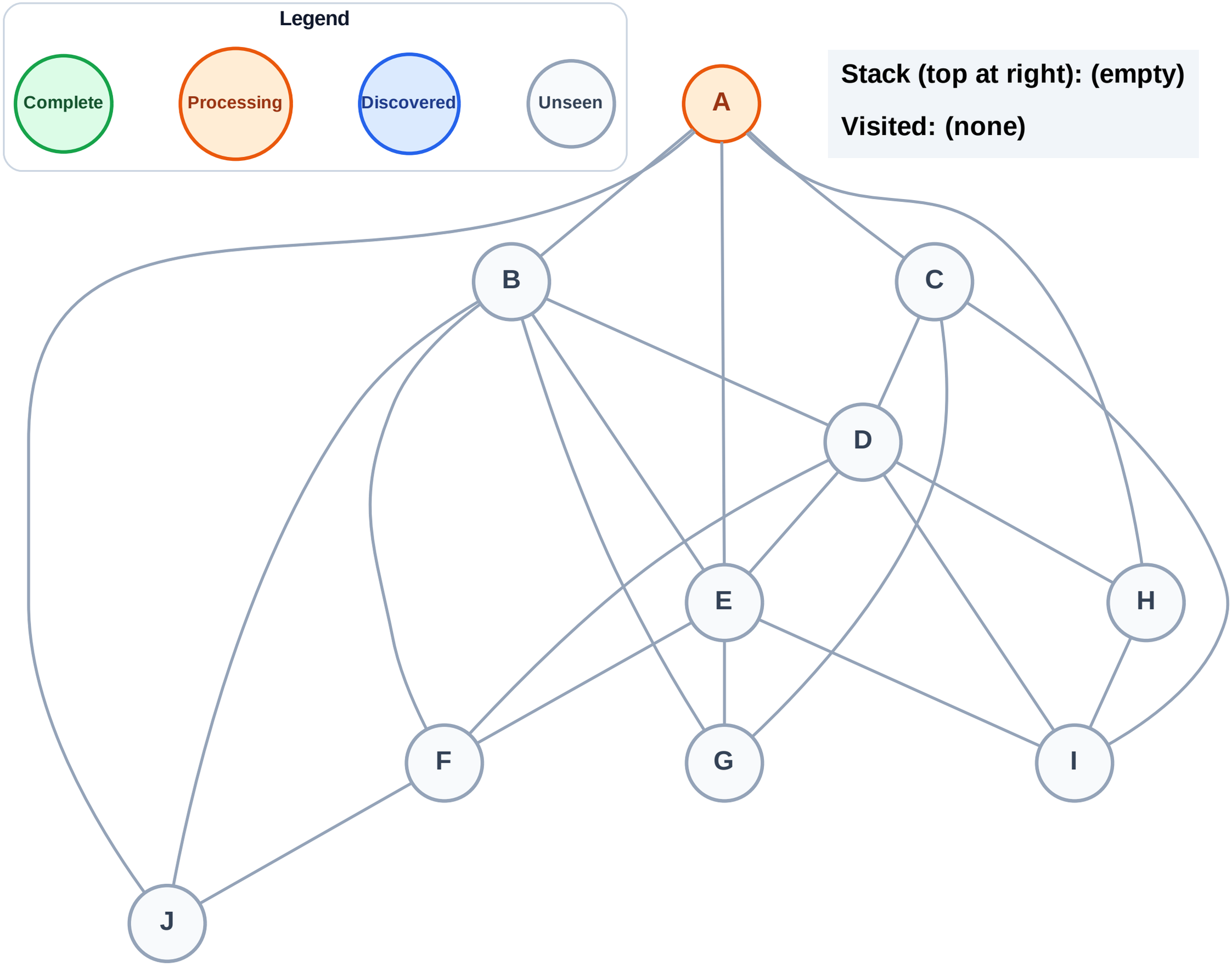
DFS Traversal

Step 1: Start at A



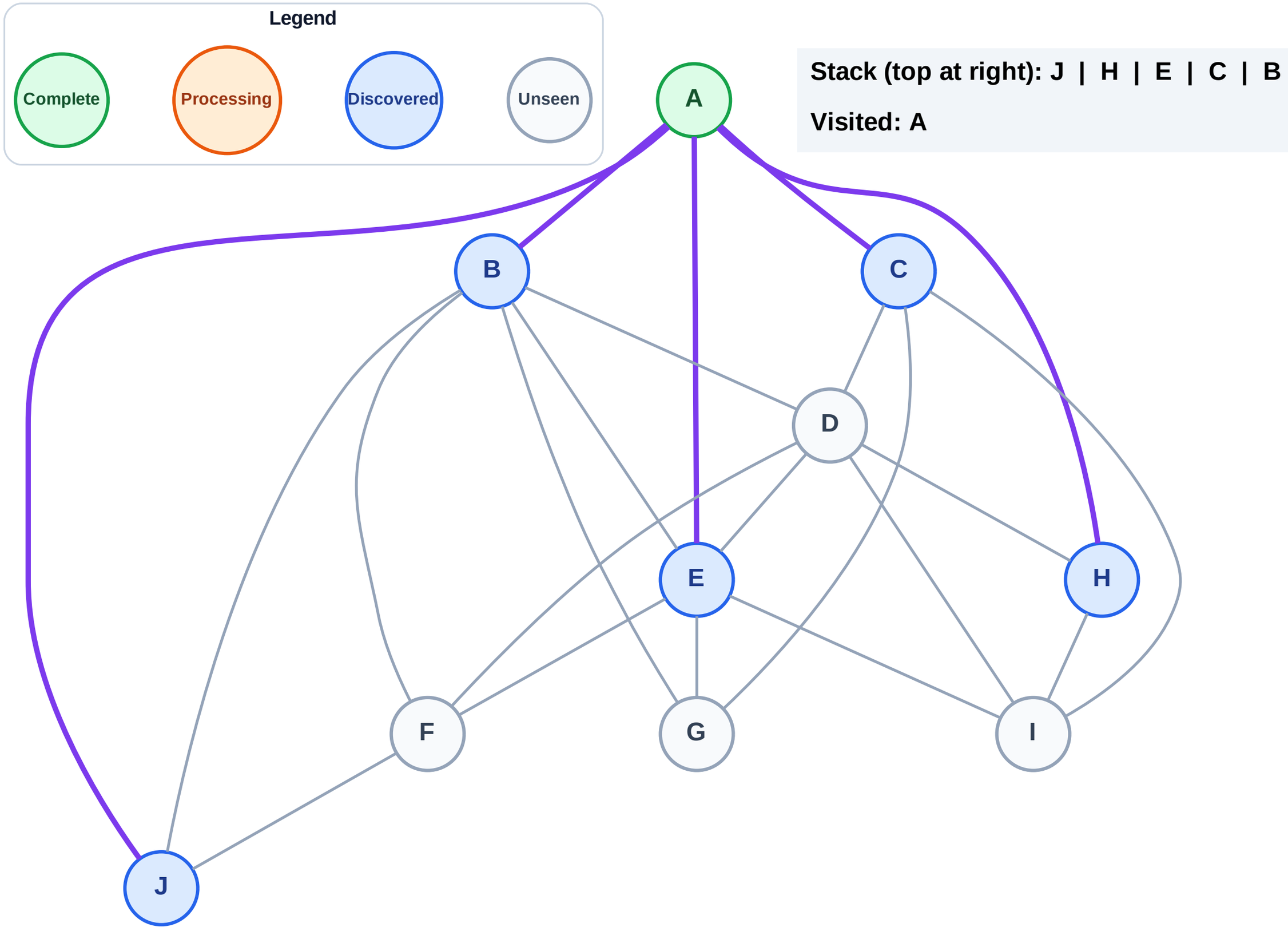
DFS Traversal

Step 2: Process node A



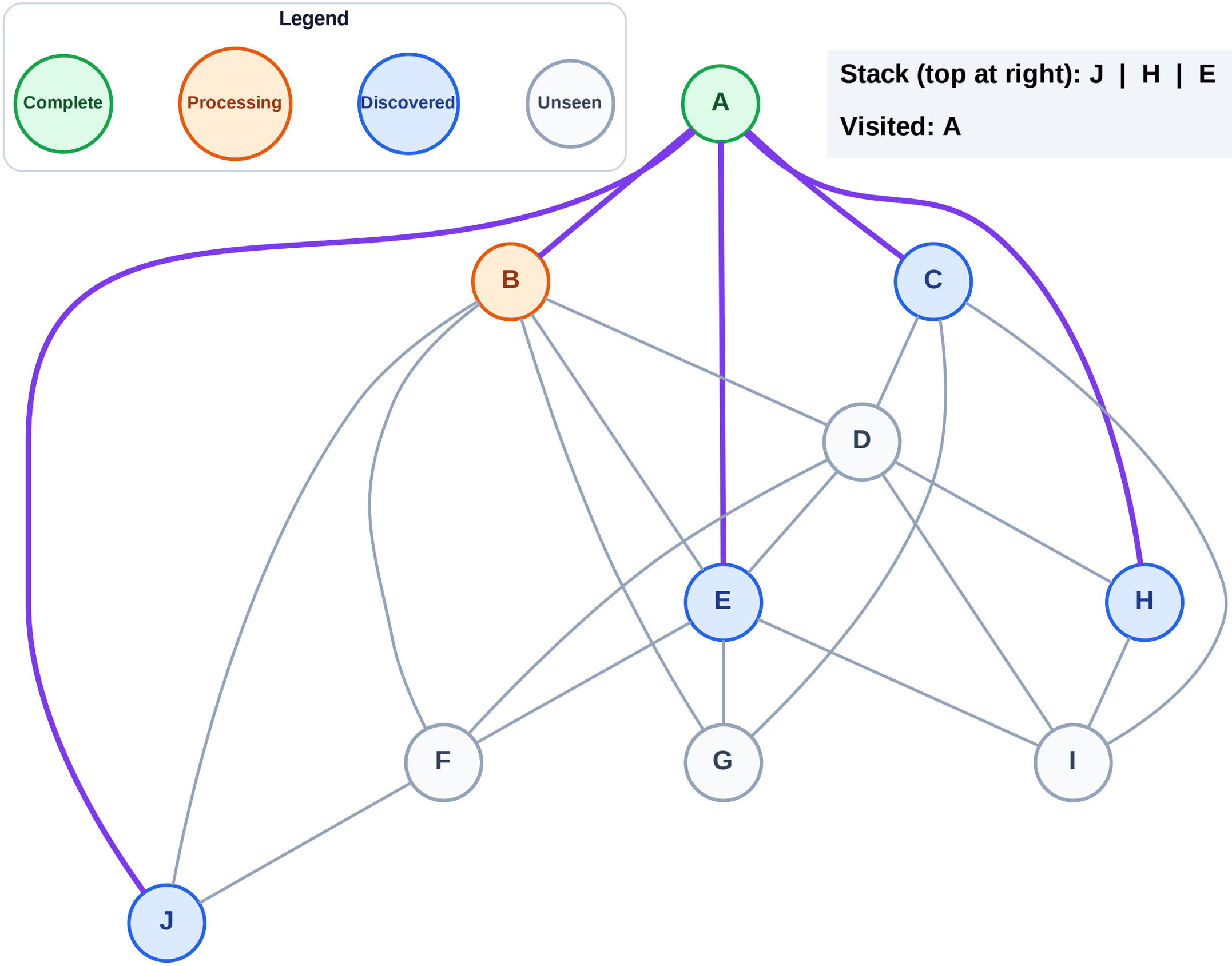
DFS Traversal

Step 3: Discover J, H, E, C, B from A



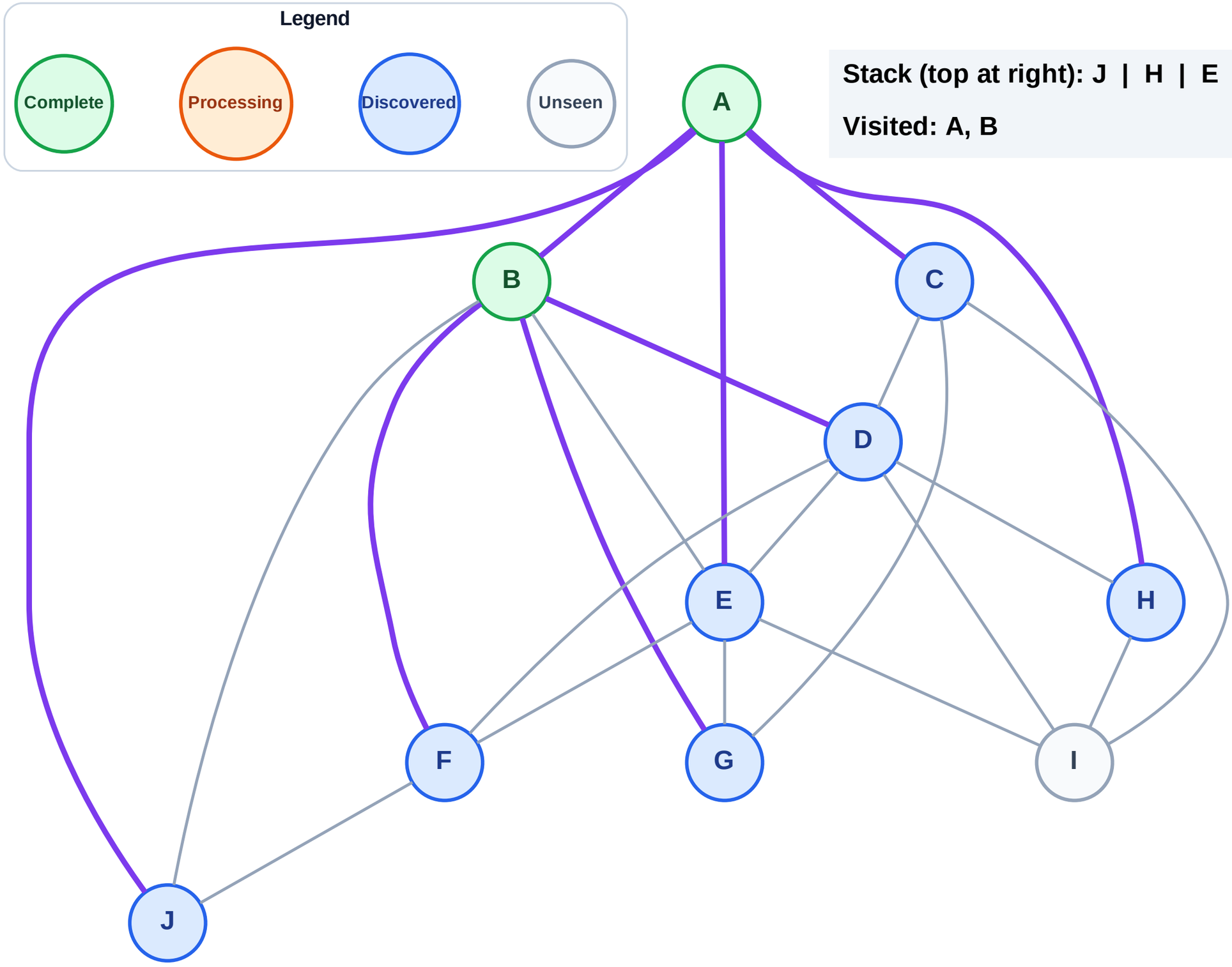
DFS Traversal

Step 4: Process node B



DFS Traversal

Step 5: Discover G, F, D from B

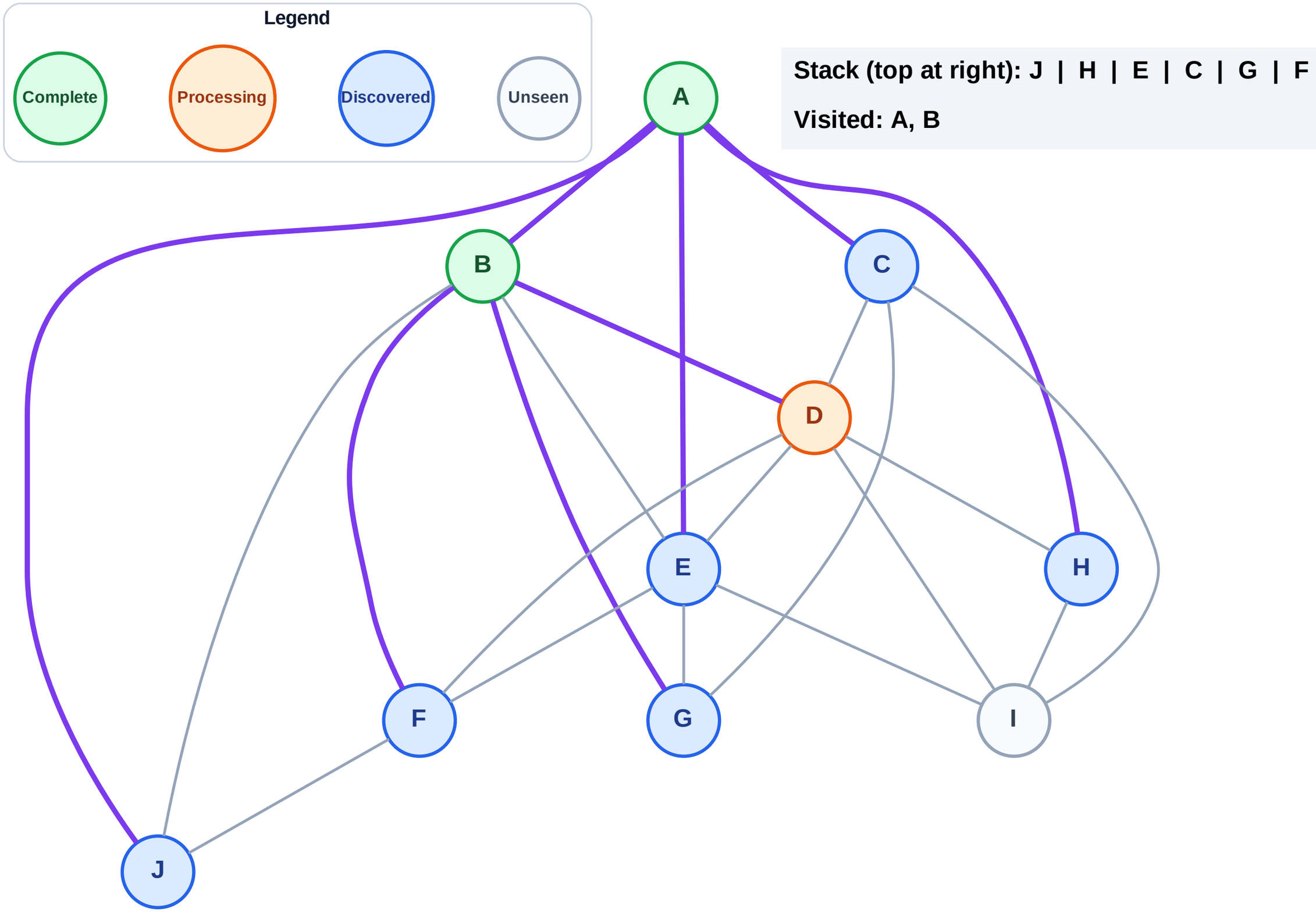


Stack (top at right): J | H | E | C | G | F | D

Visited: A, B

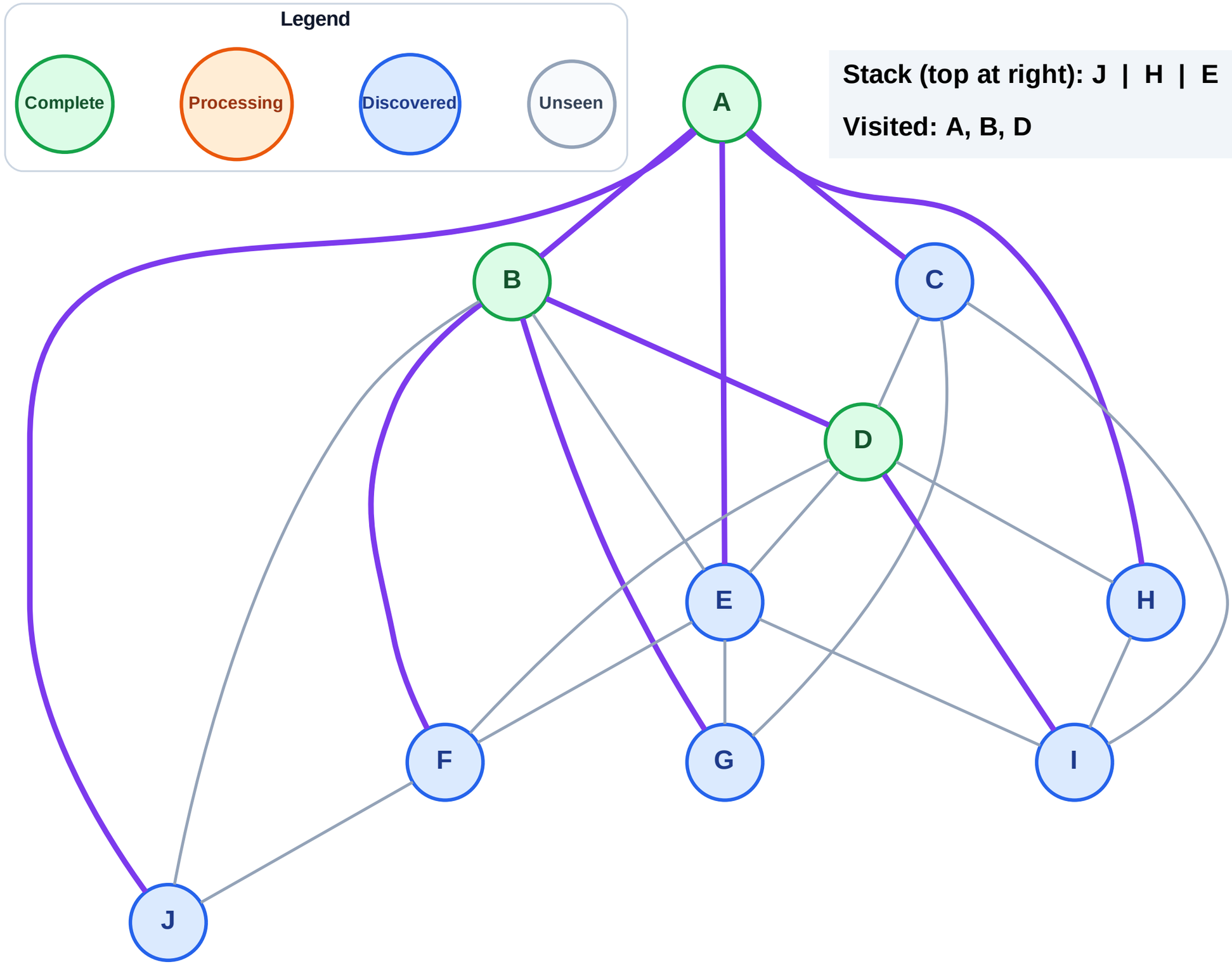
DFS Traversal

Step 6: Process node D



DFS Traversal

Step 7: Discover I from D

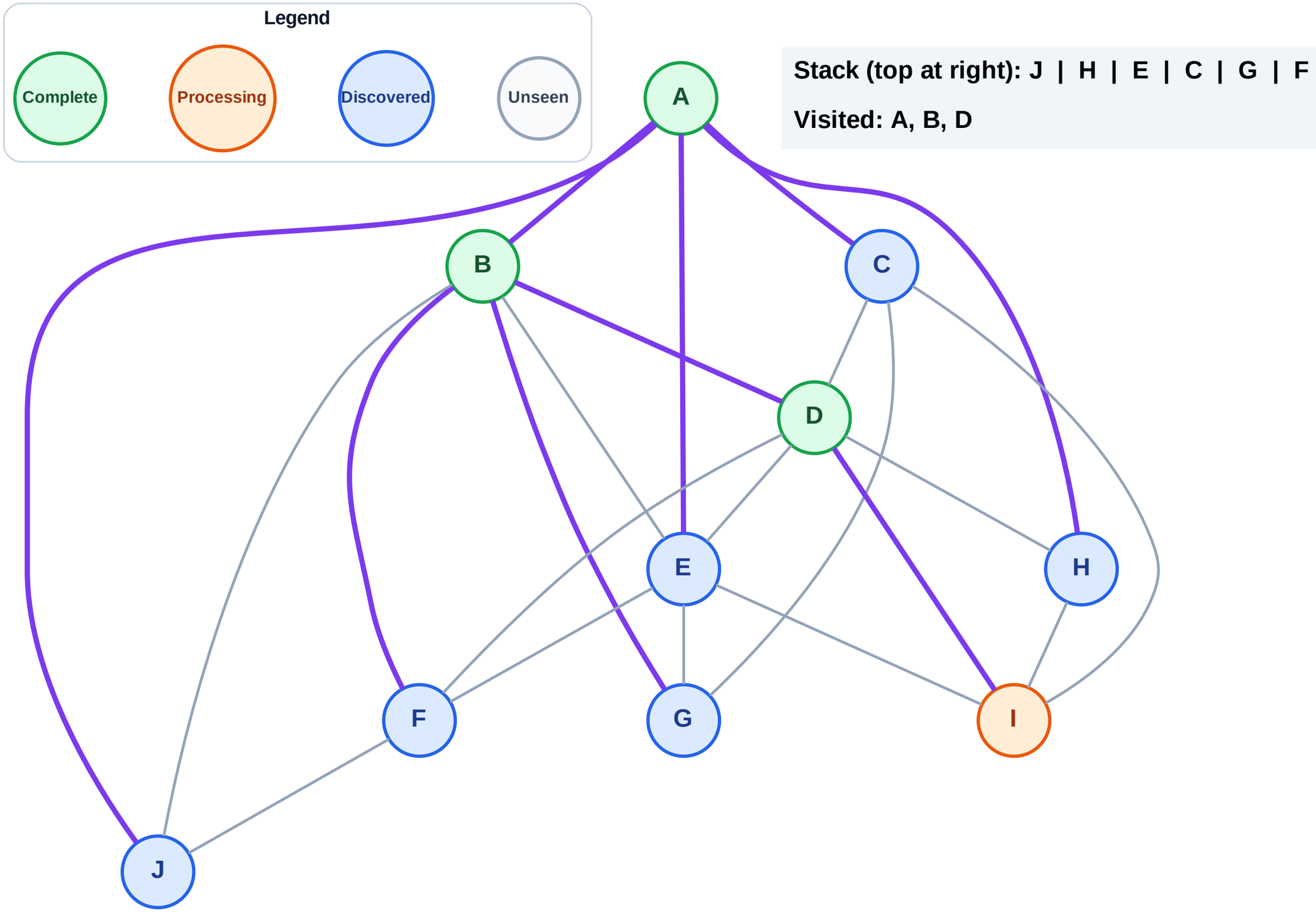


Stack (top at right): J | H | E | C | G | F | I

Visited: A, B, D

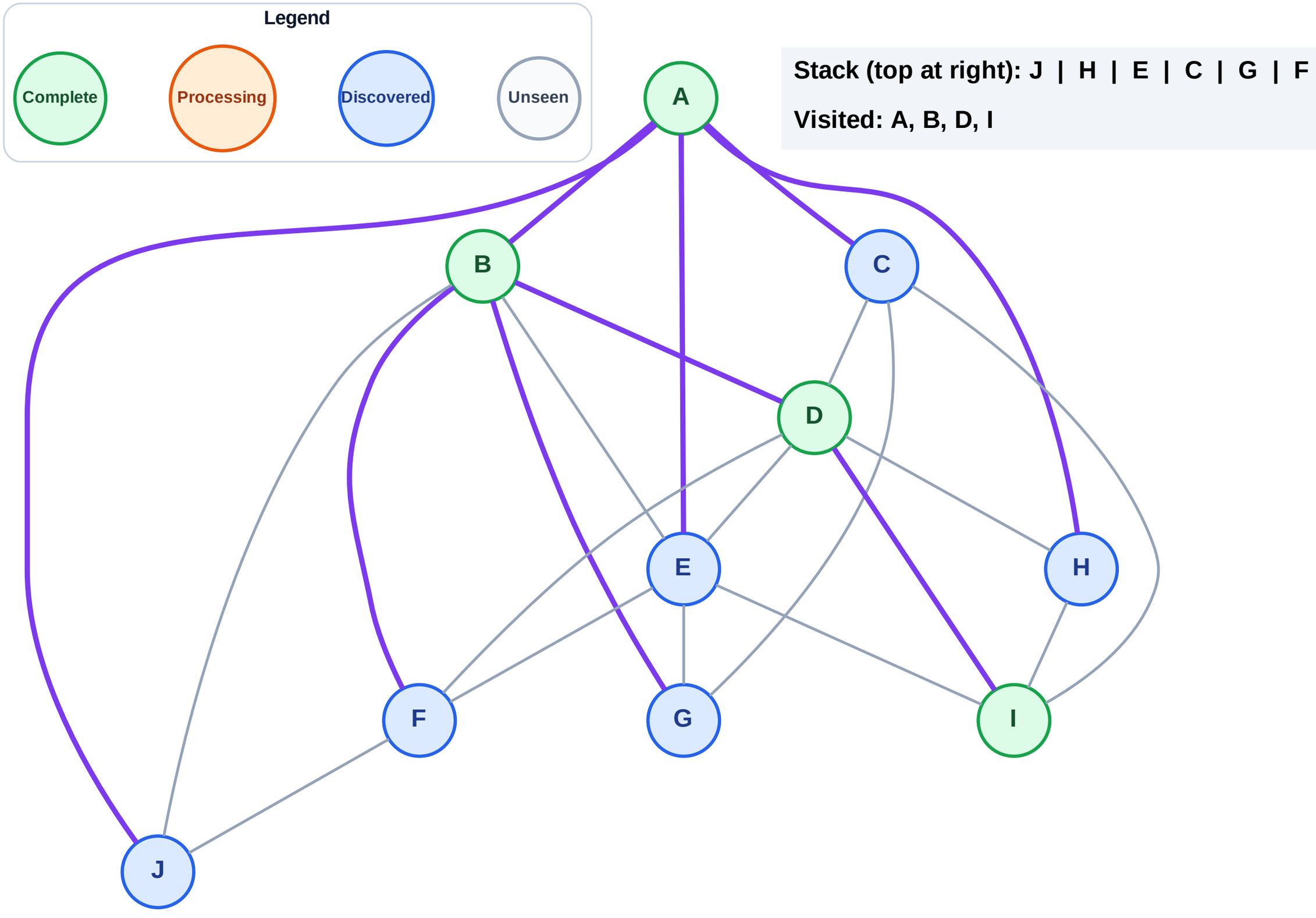
DFS Traversal

Step 8: Process node I



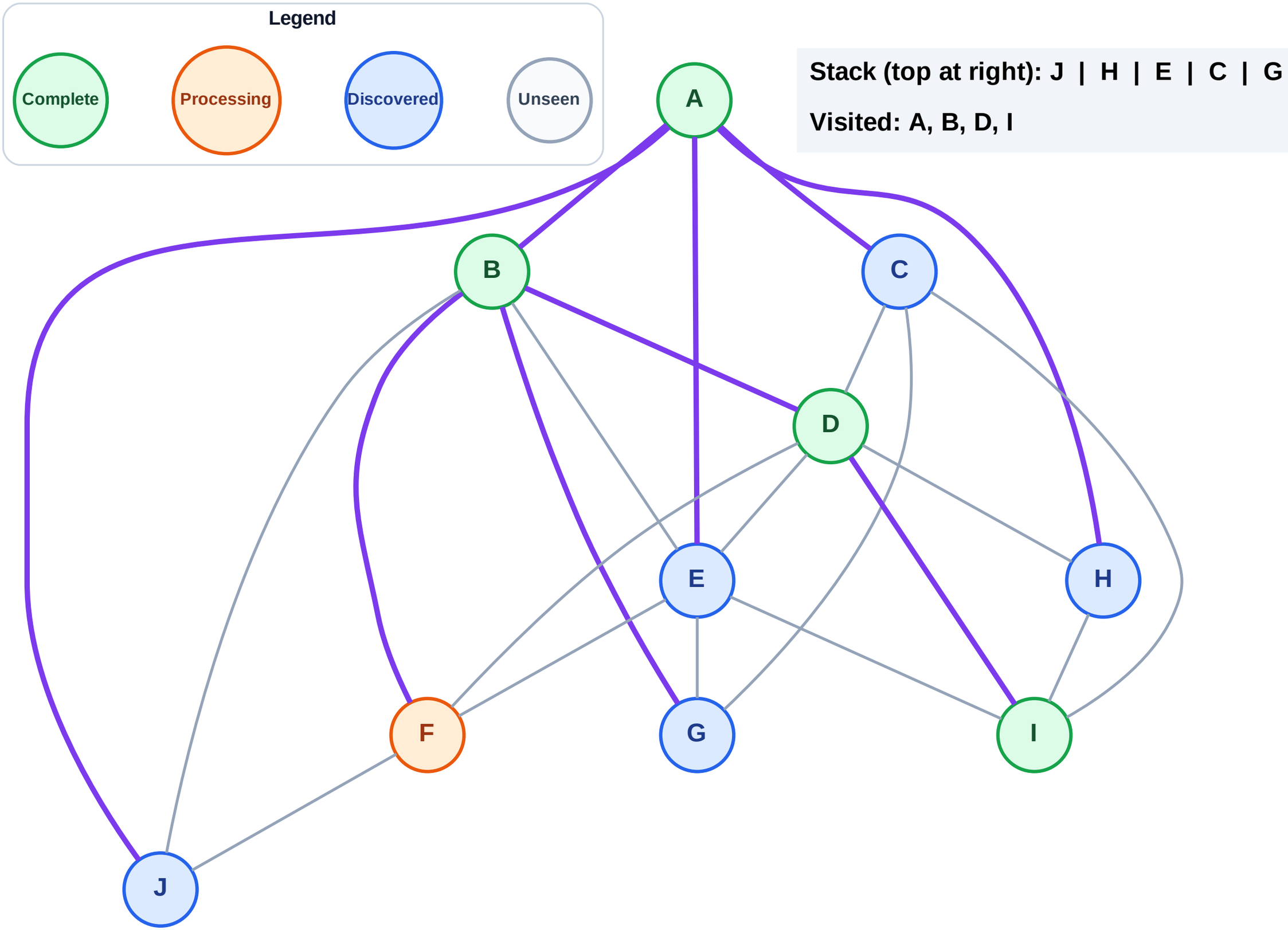
DFS Traversal

Step 9: No new neighbors from I



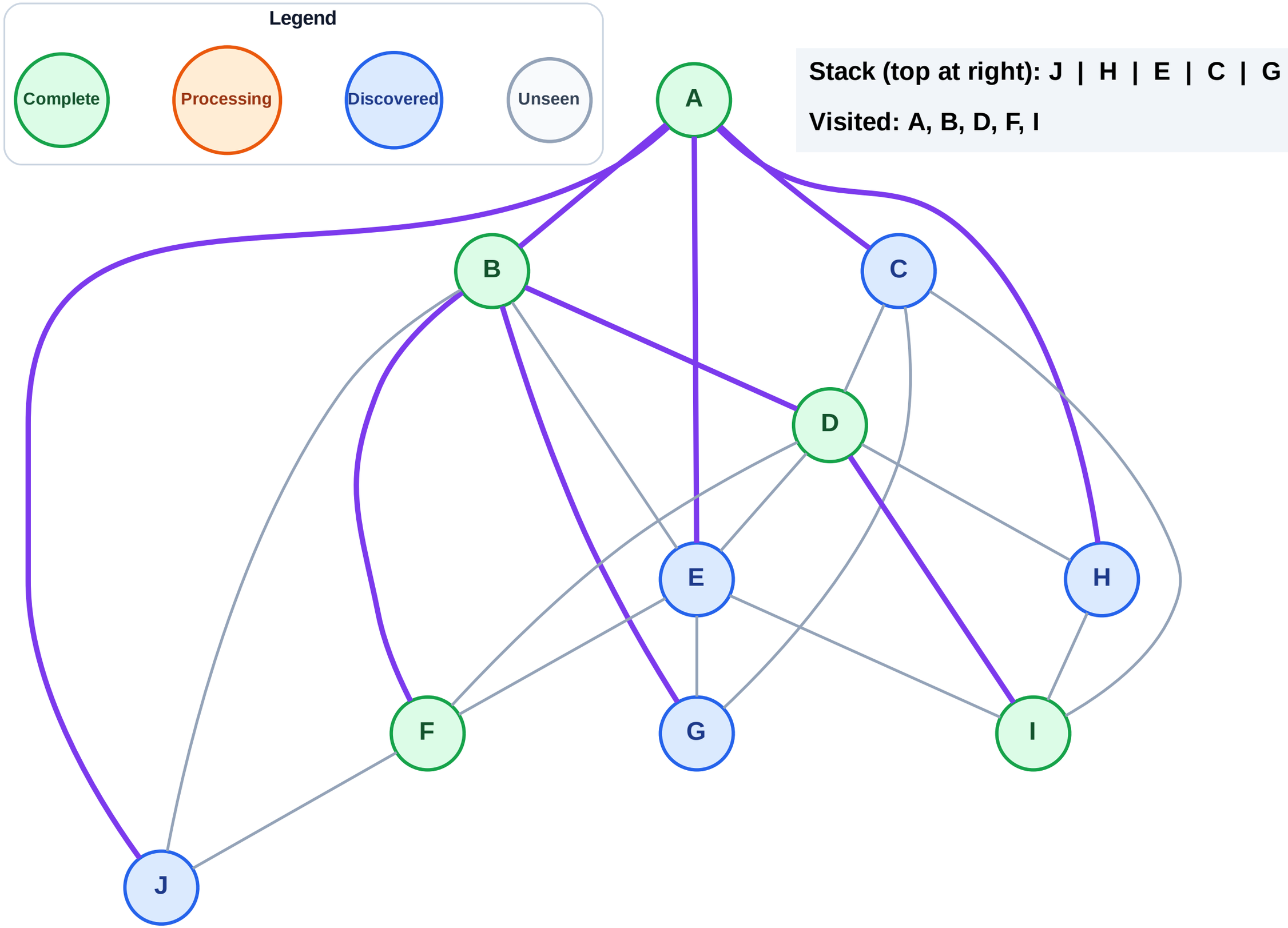
DFS Traversal

Step 10: Process node F



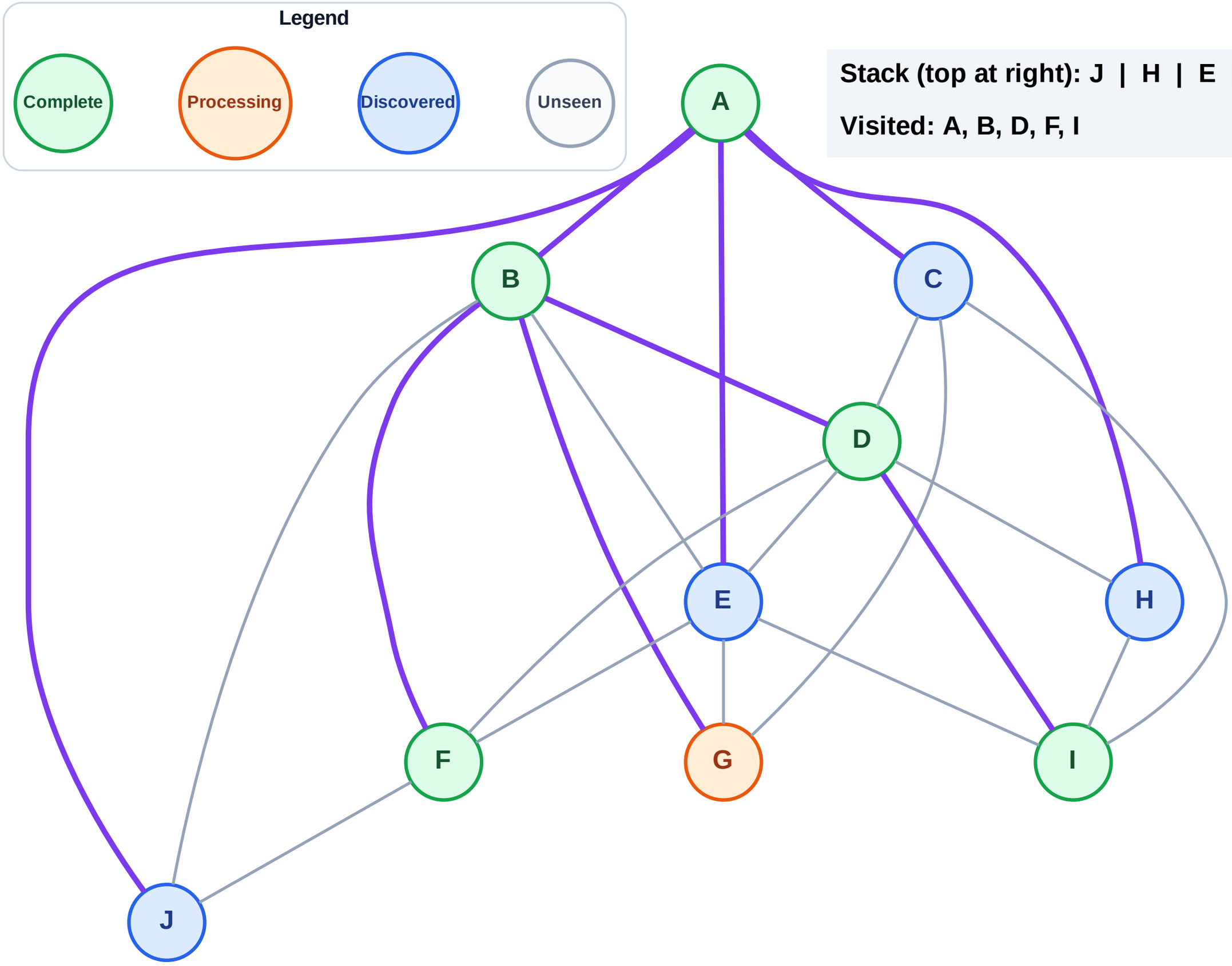
DFS Traversal

Step 11: No new neighbors from F



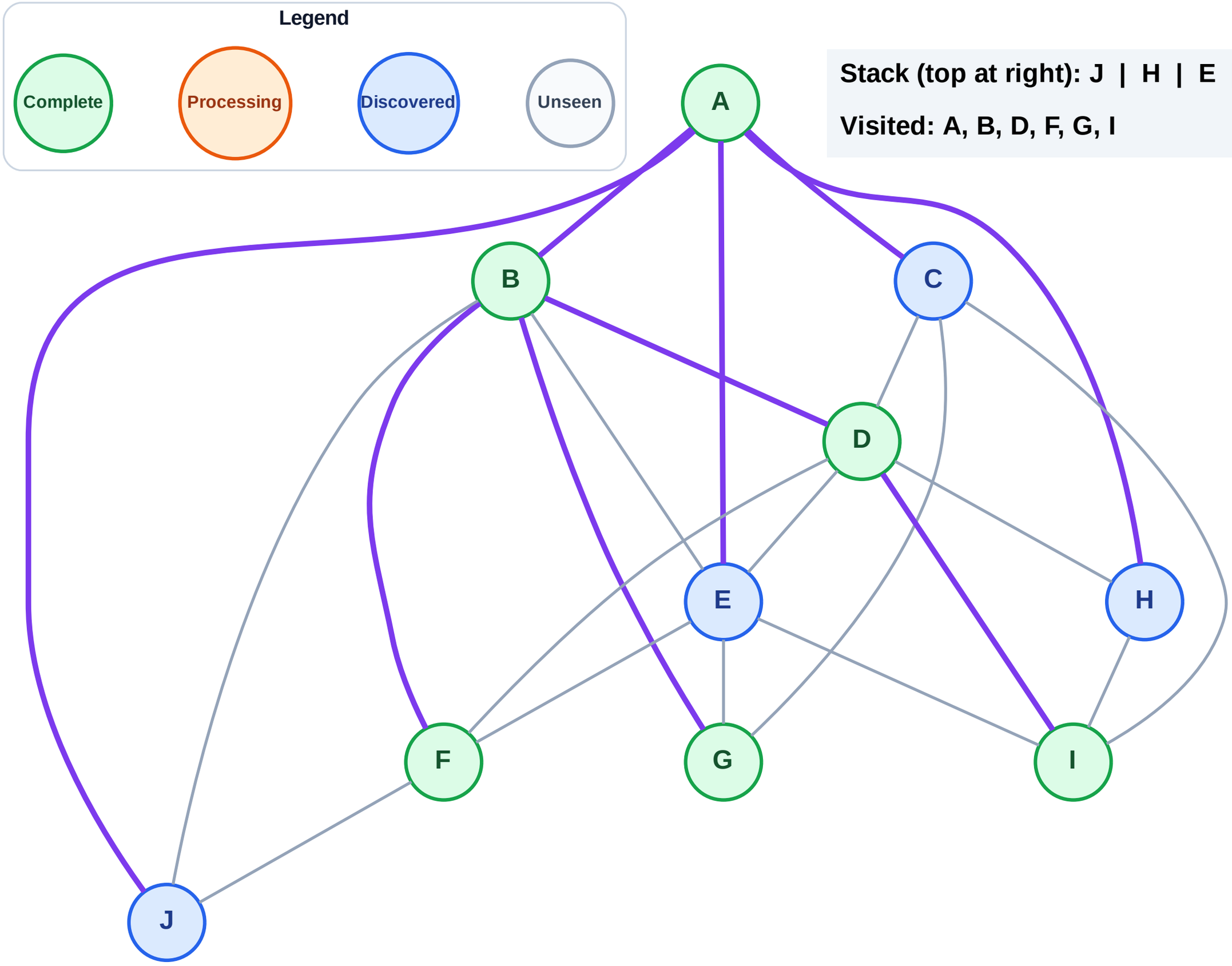
DFS Traversal

Step 12: Process node G



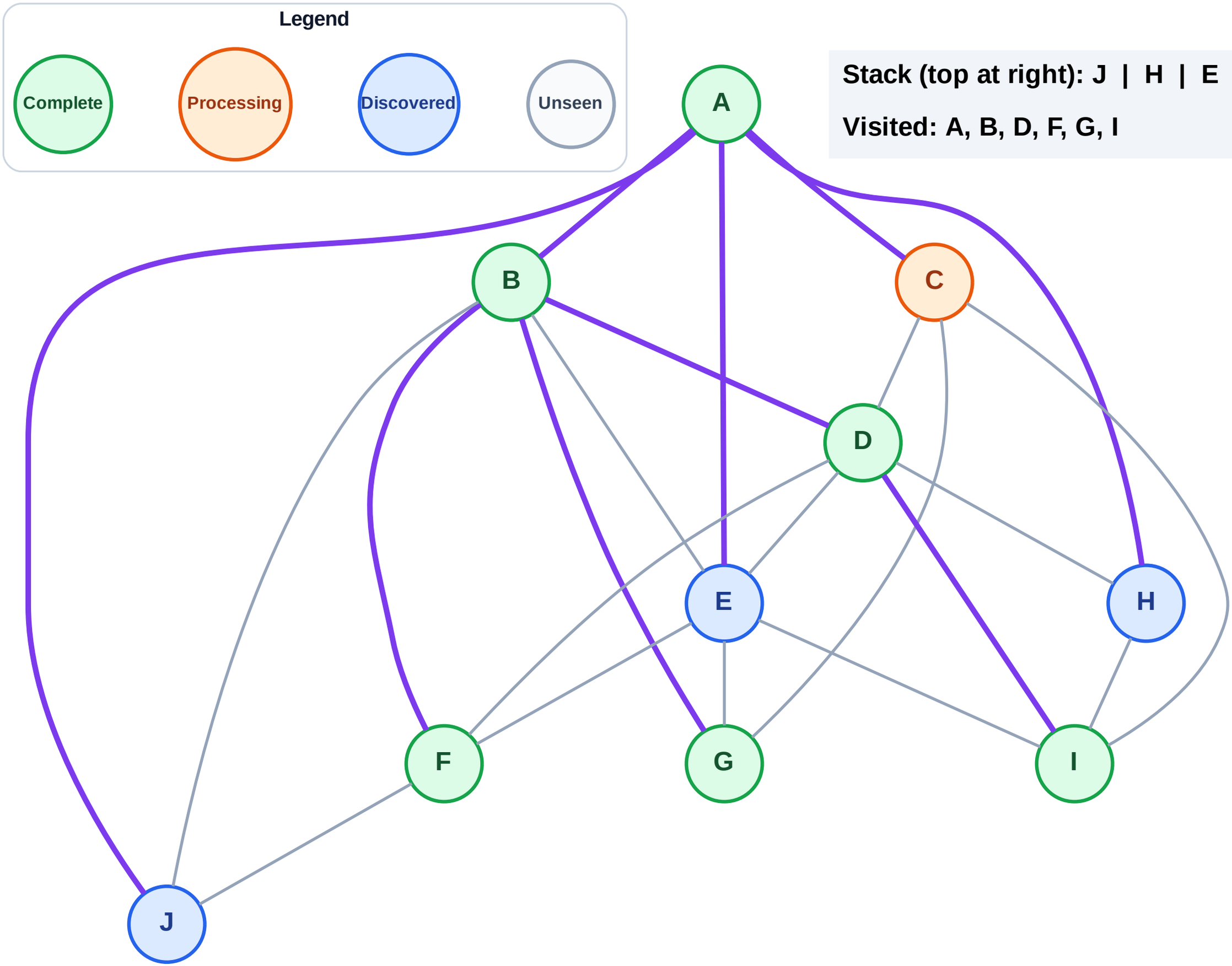
DFS Traversal

Step 13: No new neighbors from G



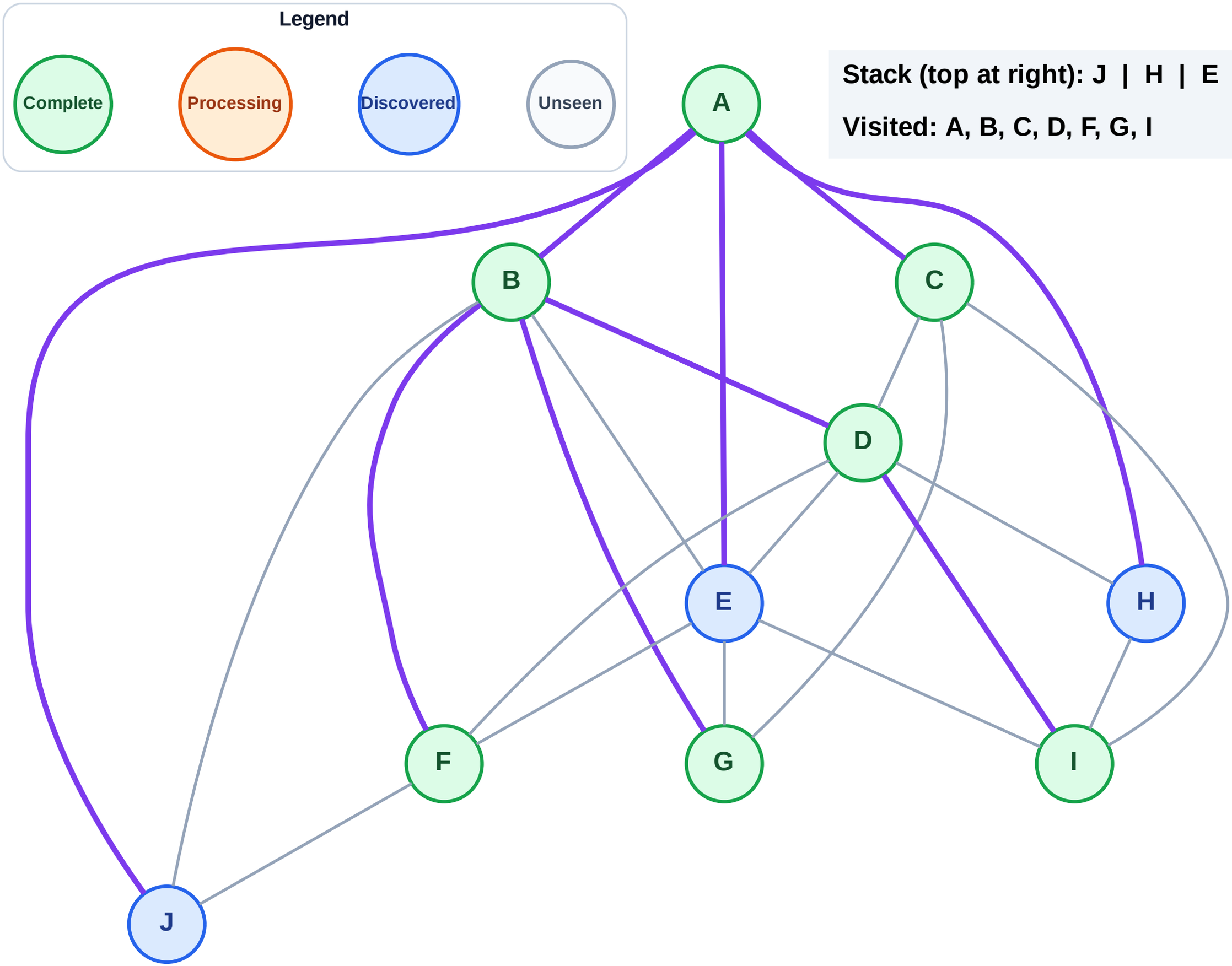
DFS Traversal

Step 14: Process node C



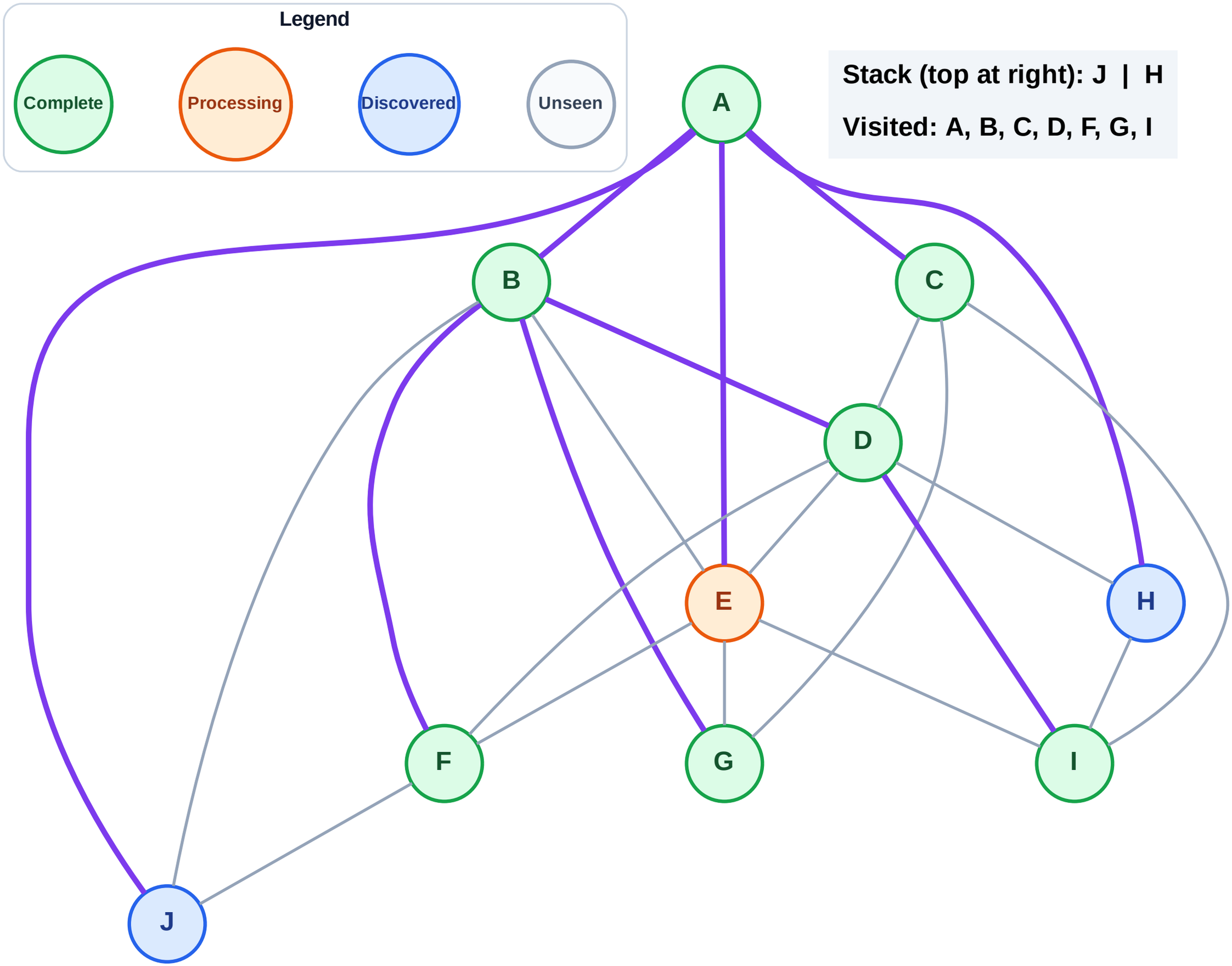
DFS Traversal

Step 15: No new neighbors from C



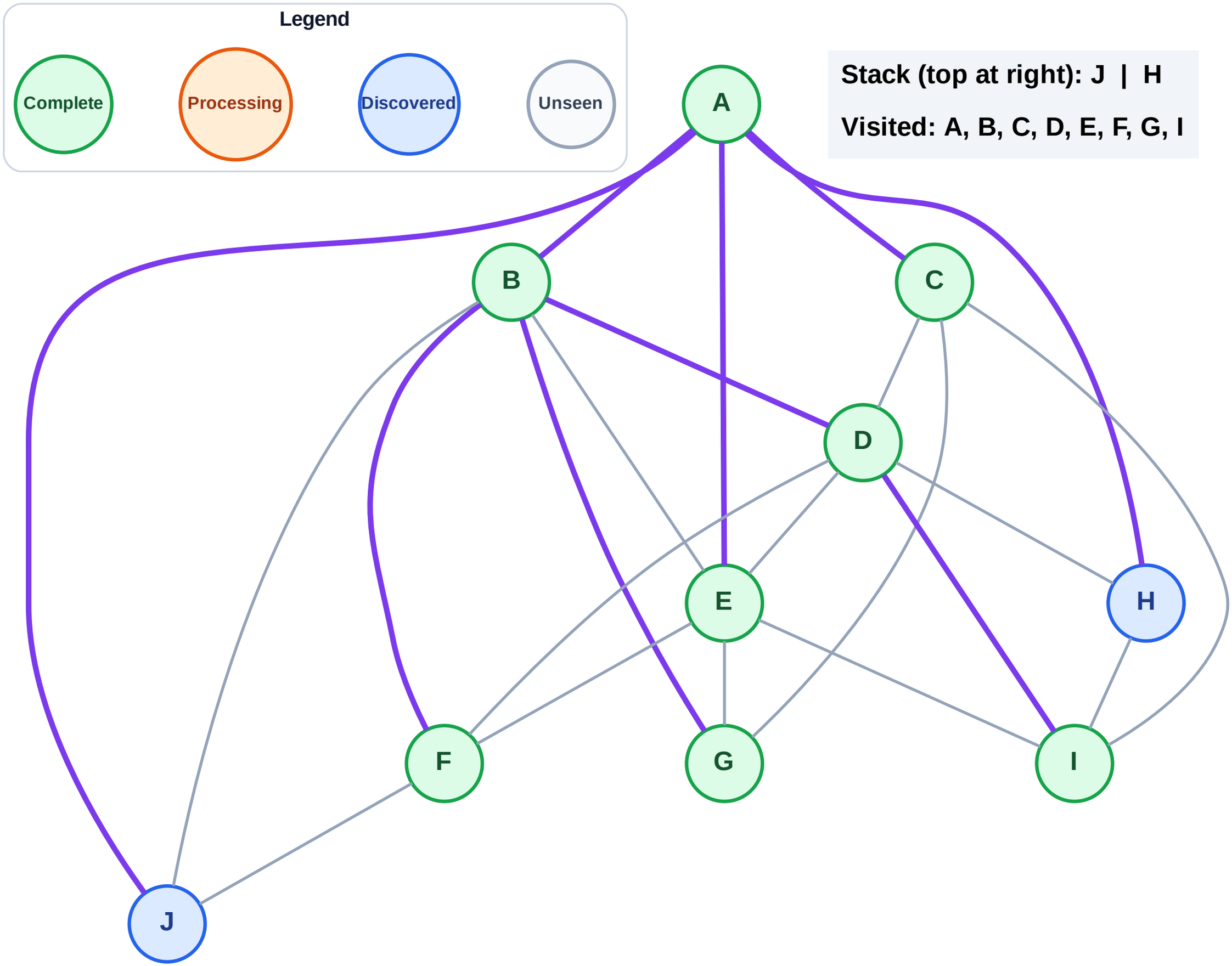
DFS Traversal

Step 16: Process node E



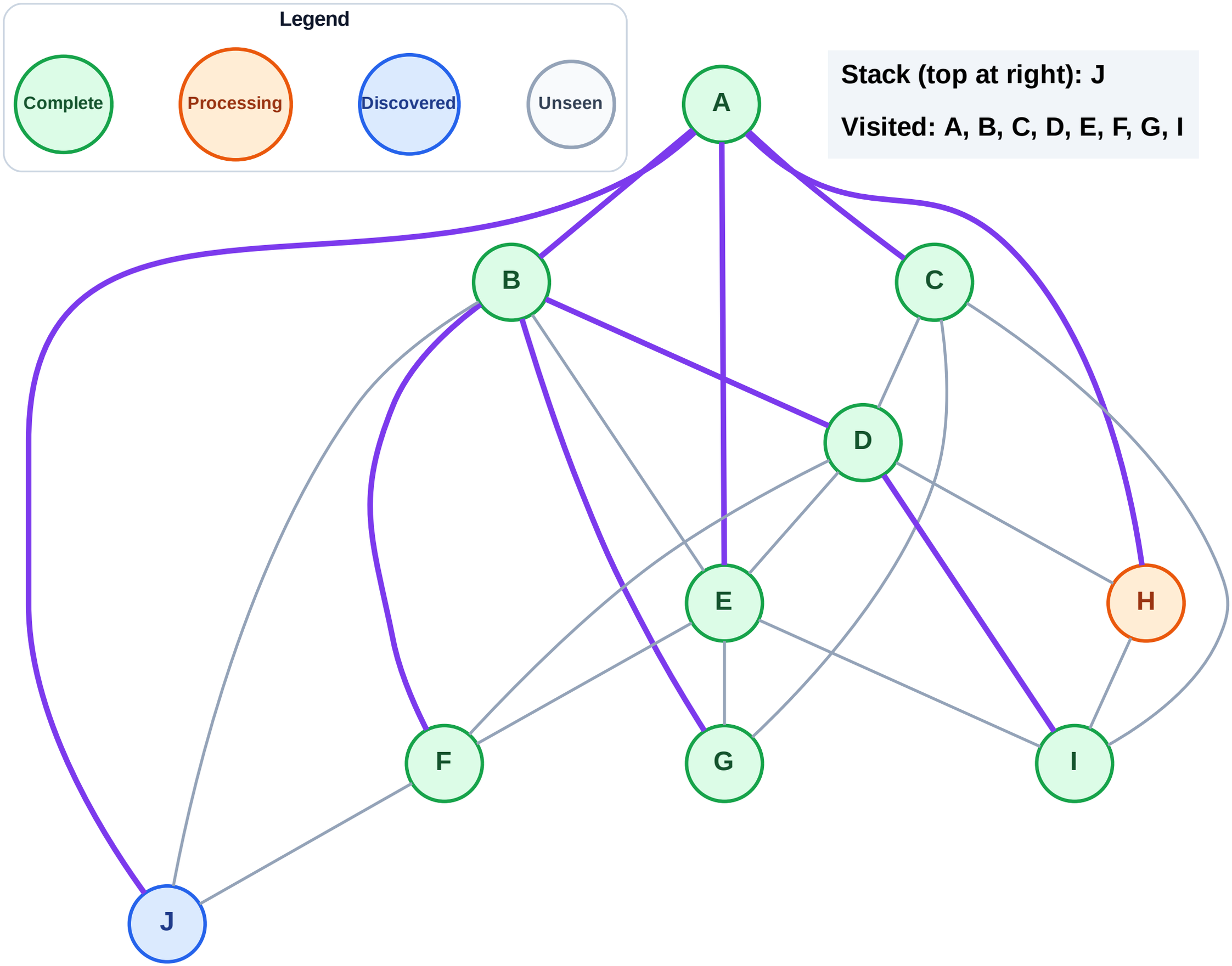
DFS Traversal

Step 17: No new neighbors from E



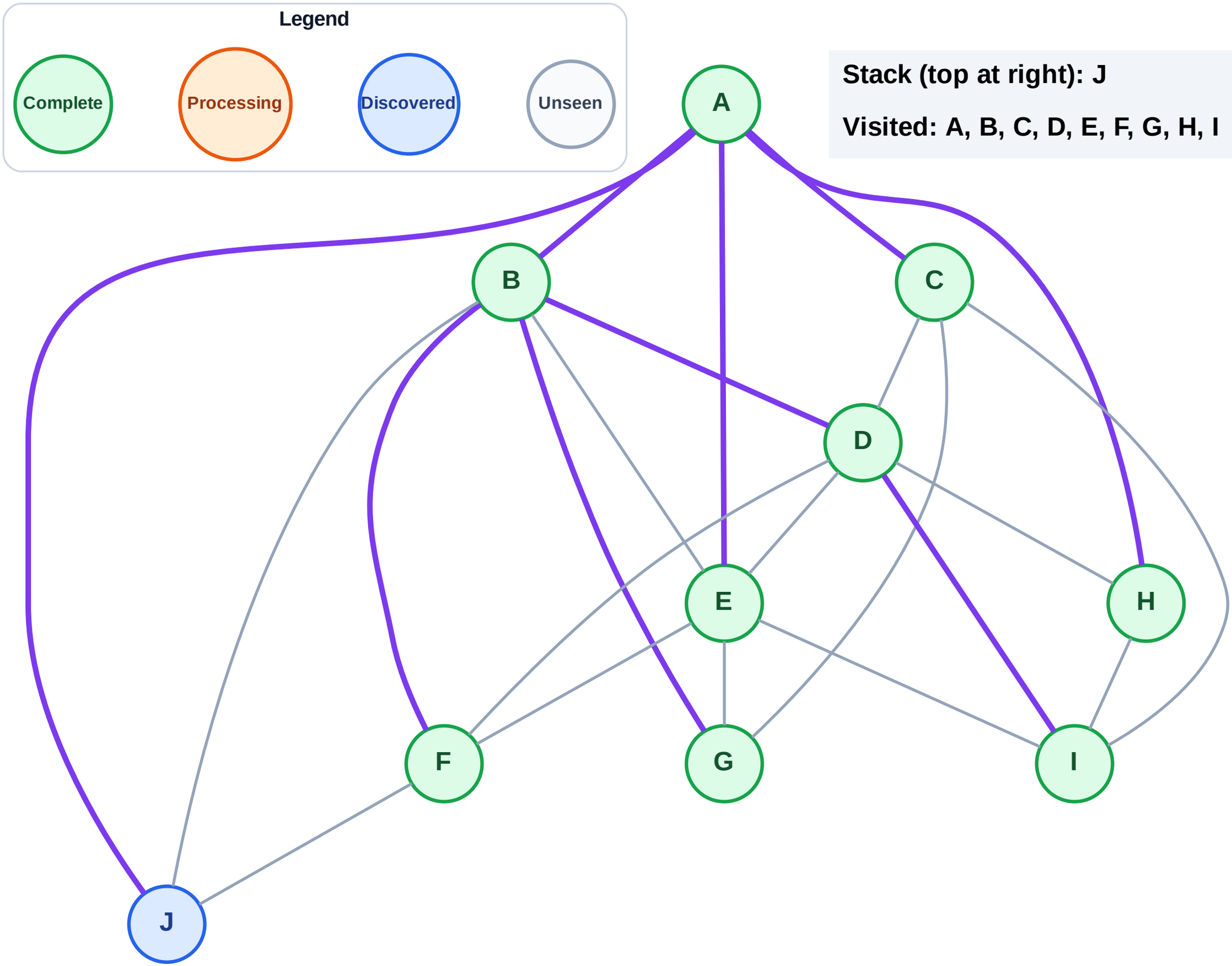
DFS Traversal

Step 18: Process node H



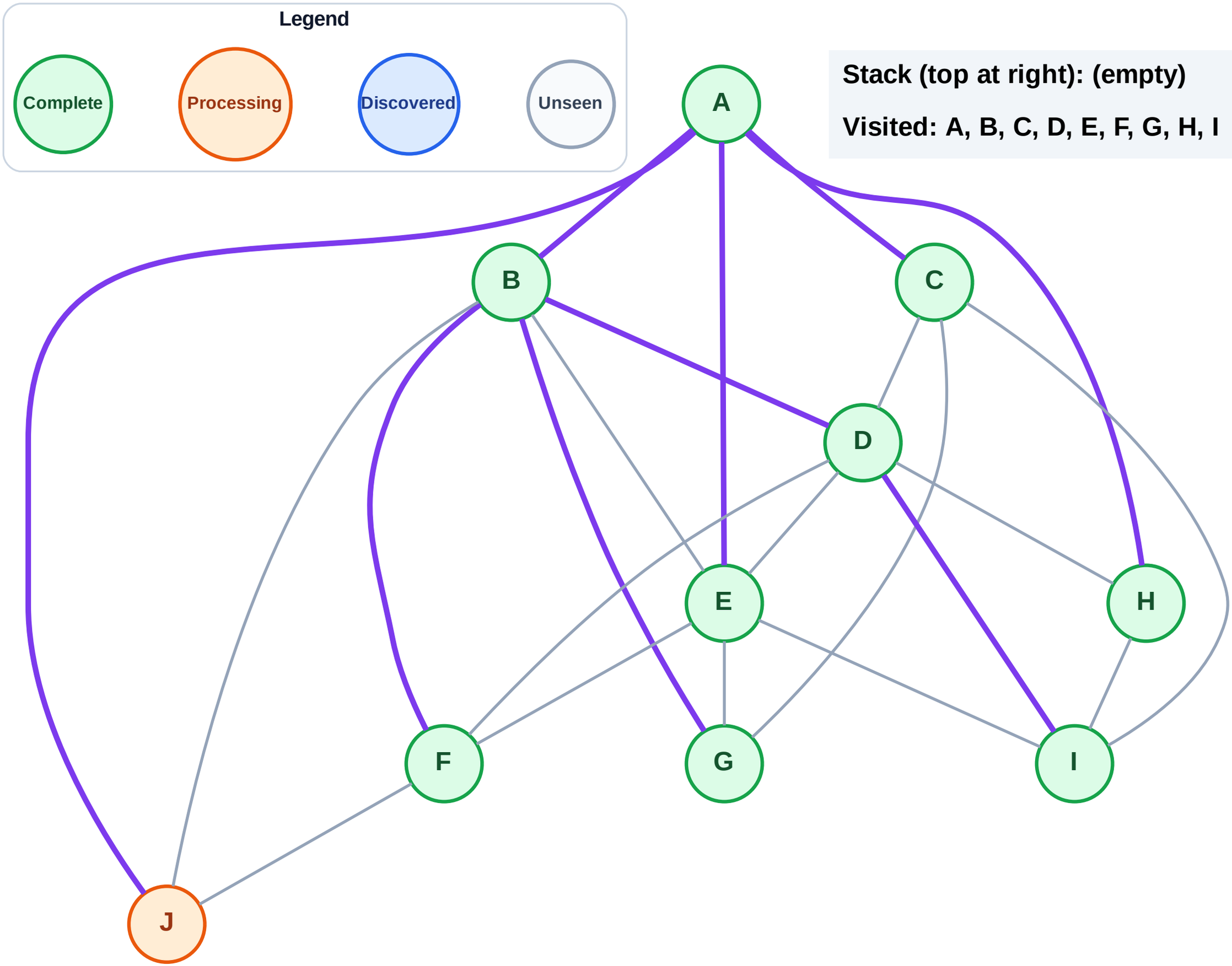
DFS Traversal

Step 19: No new neighbors from H



DFS Traversal

Step 20: Process node J



DFS Traversal

Step 21: No new neighbors from J

